Advanced Probability

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Chapter 1

Elements of Measure Theory

1.1 σ-Algebra and Measure

Let Ω be a non-empty set.

Definition 1.1.1. The set containing all subsets of Ω is defined as

$$\mathcal{P}(\Omega) \ = \ \{A \mid A \subset \Omega\},$$

Namely, the power set of Ω. (sometimes the notation of 2^{Ω} is also used).

Remark 1.1.2. *a*) Obviously $\varnothing \in \mathcal{P}(\Omega)$

b) For $A \in \mathcal{P}(\Omega)$, define

$$A^c \ = \ \{\omega \in \Omega \mid \omega \notin A\} \ \in \ \mathcal{P}(\Omega)$$

as the compliment set of A.

c) For $A, B \in \mathcal{P}(\Omega)$, define

$$A \backslash B \ = \ \{\omega \in \Omega \mid \omega \in A \text{ and } \ \omega \notin B\} \ \in \ \mathcal{P}(\Omega)$$

as B's relative complement to A.

d) Let$I$ be a index set (For example $I = \{1, \ldots, n\}, \ I = \mathbb{N}, \ I = \mathbb{R}$ or any other non-empty set) and let $A_i \subset \Omega$ where $i \in I$. Define

$$\bigcup_{i \in I} A_i = \{\omega \in \Omega \mid \exists i_0 \in I \text{ such that } \omega \in A_{i_0}\}$$

and

$$\bigcap_{i \in I} A_i \ = \ \{\omega \in \Omega \mid \omega \in A_i, \ \forall i \in I\}.$$

Theorem 1.1.3. (Distributive Laws and De Morgan Laws) *Let I be an index set and for each $i \in I$, $A_i \subset \Omega$. Then*

(1) for any $B \subset \Omega$,

$$\bigcup_{i\in I}(B \cap A_i) = B \cap \left(\bigcup_{i\in I} A_i\right) \quad \textit{and} \quad \bigcap_{i\in I}(B \cup A_i) = B \cup \left(\bigcap_{i\in I} A_i\right);$$

(2)

$$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c \quad \textit{and} \quad \left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c.$$

Definition 1.1.4. A non-empty collection of subsets of Ω, denoted by $\mathcal{A}$, is called a *σ-algebra, if*

(1) $\varnothing \in \mathcal{A}$;

(2) If $A \in \mathcal{A}$,then $A^c \in \mathcal{A}$;

(3) If $A_i \in \mathcal{A}$ $(i = 1, 2, ...)$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.

Definition 1.1.5. If $\mathcal{A}$ is a σ-algebra on Ω, then $(\Omega,\ \mathcal{A})$ is called a measurable space, and each $A \in \mathcal{A}$ is called $\mathcal{A}$-measurable.

Proposition 1.1.6. *Assume that $\mathcal{A}$ is a σ-algebra, $A_1, A_2, \ldots$ belong to $\mathcal{A}$, and $m \in \mathbb{N}$. Then, each of the following sets*

$$\varnothing\ ,\ A_1 \backslash A_2\ ,\ \bigcup_{j=0}^{m} A_j\ ,\ \bigcap_{j=0}^{m} A_j\ ,\ \bigcap_{j\in\mathbb{N}} A_j$$

also belongs to $\mathcal{A}$.

Proof. The set

$B_k := \begin{cases} A_k\ , & k \leqslant m, \\ A_m\ , & k > m, \end{cases}$ so $B_k \in \mathcal{A}$, $k \in \mathbb{N}$ and

$$\bigcup_{k\in\mathbb{N}} B_k = \bigcup_{j=0}^{m} A_j\ \in \mathcal{A}\ .$$

The remaining conclusions can be directly derived using De Morgan's laws. □

Let ' s now look at a few examples.

Example 1.1.7. a) *$\mathcal{P}(\Omega)$ is a σ-algebra on Ω*

b) *$\{\Omega,\ \varnothing\}$ is the trivial σ-algebra on Ω.*

c) Let Ω be a set such that $\sharp(\Omega) = \infty$, where $\sharp(\Omega)$ denotes the number of elements in Ω. Then, the collection $\{A \subset \Omega : A$ *or* A^C is countable$\}$ *is a σ-algebra on Ω.*

d) *The collection $\mathcal{F} = \{\varnothing,\ \{a\},\ \{b\},\ \{c,\ d\},\ \{a,\ c,\ d\},\ \{b,\ c,\ d\},\ \Omega\}$* is not a *$\sigma$-algebra on $\Omega = \{a,\ b,\ c,\ d\}$*, because $\{a,\ b\} \notin \mathcal{F}$.

Proposition 1.1.8. (Intersection of σ-algebra) *Let I be an index set, and for each $i \in I$, let $\mathcal{A}_i$ be a σ-algebra on Ω. Then*

$$\bigcap_{i\in I}\mathcal{A}_i \;=\; \{A \;\subset\; \Omega \mid A \in A_i,\; \forall i \in I\}$$

is also a σ-algebra on Ω.

Proof. Left as an exercise. □

As an application of the previous proposition, we have the following points.

Theorem 1.1.9. *If we have a subset $\mathcal{C}$ of Ω, define*

$$(^*) \qquad\qquad \mathcal{A} := \bigcap_{\mathcal{C}\subset\mathcal{G},\ \mathcal{G}\ \sigma-algebra} \mathcal{G},$$

as the intersection of all σ-algebras containing $\mathcal{C}$. Then $\mathcal{A}$ is the smallest σ-algebra containing $\mathcal{C}$, meaning that for every σ-algebra$\mathcal{A}'$ containing $\mathcal{C}$, we have $\mathcal{A} \subset \mathcal{A}'$. We denote this situation as $\mathcal{A} = \sigma(\mathcal{C})$, and it is called the σ-algebra generated by $\mathcal{C}$.

Proof. By Proposition 1.1.8, $\mathcal{A}$ is a σ-algebra. Since $\mathcal{G} = \mathcal{P}(\Omega)$ is a candidate for the intersection under $(^*)$, and $(^*)$ has a well-defined meaning (not an empty intersection). Suppose $\mathcal{A}'$ is a σ-algebra such that $\mathcal{C} \subset \mathcal{A}'$. Then, $\mathcal{A}'$ is obviously contained in the intersection under $(^*)$, so $\mathcal{A}' \supset \mathcal{A}$. □

Since $\mathcal{A}$ is the smallest in $(^*)$, $\sigma(\mathcal{C}) = \mathcal{A}$ is a valid notation.

Definition 1.1.10. Consider $\Omega = \mathbb{R}$ and

$$\mathcal{E} \;:=\; \{(a, b] \mid a \in \mathbb{R},\; b \in \mathbb{R},\; a \leqslant b\}. \qquad (\text{Note: } \varnothing = (a, a] \in \mathcal{E})$$

Define

$$\mathcal{B}(\mathbb{R}) \;:=\; \sigma(\mathcal{E})$$

and call it the Borel σ-algebra on $\mathbb{R}$.

Proposition 1.1.11. *We have*

- $\varnothing \in \mathcal{B}(\mathbb{R}), \mathbb{R} \in \mathcal{B}(\mathbb{R})$
- $\{c\} \in \mathcal{B}(\mathbb{R}), \quad \forall c \in \mathbb{R}$
- *for all $a \in \mathbb{R}, b \in \mathbb{R}$ and $a < b$,*

$$[a,\ b]\ ,\ [a,\ b),\ (a,\ b]\ ,\ (a,\ b)\ \in \mathcal{B}(\mathbb{R})$$

- $\mathbb{N} \in \mathcal{B}(\mathbb{R}), \mathbb{Q} \in \mathcal{B}(\mathbb{R})$.

If we take the complement, countable union, or countable intersection of the above sets, the resulting sets will still be contained in $\mathcal{B}(\mathbb{R})$.

Definition 1.1.12. LetΩ be a set and $\mathcal{A}$ be a σ-algebra on Ω. A measure on $(\Omega,\ \mathcal{A})$ is a function $\mu : \mathcal{A} \to [0,\ \infty]$ The measure satisfies the following properties:

(1) $\mu(\varnothing) = 0$;

(2) If $A_i \in \mathcal{A}, i = 1$, 2, . . ., are pairwise disjoint, then

$$\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)\ .$$

Here, "A_i are pairwise disjoint" means that for any $i \neq j,\ A_i \cap A_j = \varnothing$. This property is also known as countable additivity.

Definition 1.1.13. If$\mu$is a measure on $(\Omega,\ \mathcal{A})$, we call $(\Omega,\ \mathcal{A},\ \mu)$ a measure space.

Definition 1.1.14. *Let* $(\Omega,\ \mathcal{A},\ \mu)$ be a measure space.

a) μ is called a finite measure, *if* $\mu(\Omega) < \infty$;

b) μ is called a probability measure, *if* $\mu(\Omega)\ =\ 1$*;*

c) μ is called a σ-finite measure, *if there exist sets* $A_i \in \mathcal{A}, i = 1$, 2, . . ., such that

$$\bigcup_{n=1}^{\infty} A_n = \Omega \quad \text{and} \quad \mu(A_n) < \infty, \quad \forall n \in \mathbb{N}.$$

Now, let's look at a few examples.

Example 1.1.15. (a) Fix $a \in \Omega$, define

$$\delta_a(A) := \begin{cases} 1\ , & a \in A\ , \\ 0\ , & a \notin A\ , \end{cases}$$

where $A \subset \Omega$. Then δ_a: $\mathcal{P}(\Omega) \to [0,\ \infty]$ is a probability measure, and it is called the Dirac measure on Ω.

(b) For $A \subset \Omega$ define $\mu(A) := \sharp(A)$. Then $\mu : \mathcal{P}(\Omega) \to [0,\ \infty]$ is a measure. It is a finite measure (or σ-finite measure) if and only if Ω is finite (or countably infinite). This measure μ is commonly referred to as the counting measure on Ω.

(c) For $A \subset \Omega$, define $\mu(A) := 0$ if $A = \varnothing$ and $\mu(A) := \infty$ if $A \neq \varnothing$. Then $(\Omega,\ \mathcal{P}(\Omega),\ \mu)$ is a measure space.

Notation 1.1.16. Let A *and* $A_i, i = 1, 2, \cdots$ be subsets of Ω.

(i) $A_n \uparrow A$ means that $A_1 \subset A_2 \subset \ldots$ and $\bigcup_{n=1} A_n = A$.

(ii) $A_n \downarrow A$ means that $A_1 \supset A_2 \supset \ldots$ and $\bigcap_{n=1} A_n = A$.

We now collect the most important properties of measures.

Proposition 1.1.17. (Properties of a measure) *Let* $(\Omega,\ \mathcal{A},\ \mu)$ *be a measure space. For* $A, B \in \mathcal{A}$ *and* $A_i \in \mathcal{A}, i = 1, 2, \cdots$*, we have:*

(i) If A_1*, . . . ,* A_n *are pairwise disjoint, then* $\mu(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mu(A_i)$*, that is,* μ *is additive.*

(ii) $\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B)$ *.*

(iii) If $A \subset B$ *and* $\mu(A) < \infty$*, then* $\mu(B\backslash A) = \mu(B) - \mu(A)$*.*

(iv) $A \subset B \Rightarrow \mu(A) \leqslant \mu(B)$ *, namely,* μ *is monotonic.*

(v) (Continuity from below) *If* $A_n \uparrow A$*,* $\mu(A_n) \uparrow \mu(A)$*.*

(vi) (Continuity from above) *If* $A_n \downarrow A$ *and* $\mu(A_1) < \infty$*,* $\mu(A_n) \downarrow \mu(A)$*.*

(vii) $\mu(\bigcup_j A_j) \leqslant \sum_j \mu(A_j)$*, namely,* μ *is countably subadditive.*

Proof. (i) Define a set

$B_k := \begin{cases} A_k\ , & k \leqslant n, \\ \varnothing\ , & k > n, \end{cases}$ Then $B_k \in \mathcal{A}$, $k \in \mathbb{N}$ and

$$\bigcup_{k\in\mathbb{N}} B_k = \bigcup_{j=1}^{n} A_j \ \in \mathcal{A}\ .$$

As a result,

$$\mu(\bigcup_{i=1}^{n} A_i) = \mu(\bigcup_{i=1}^{\infty} B_i) = \sum_{i=1}^{\infty} \mu(B_i) = \sum_{i=1}^{n} \mu(A_i).$$

(ii) From $A \cup B = A \cup (B\backslash A)$ and $A \cap (B\backslash A) = \varnothing$, by additivity, we get

$$\mu(A \cup B) = \mu(A) + \mu(B\backslash A). \tag{1.1.1}$$

Similarly, from $B = (A \cap B) \cup (B\backslash A)$ and $(A \cap B) \cap (B\backslash A) = \varnothing$, we can conclude

$$\mu(A \cap B) + \mu(B\backslash A)\ = \mu(B). \tag{1.1.2}$$

By adding equations (1.1.1) and (1.1.2), we obtain

$$\mu(A \cup B) + \mu(A \cap B) + \mu(B\backslash A)\ = \mu(A) + \mu(B) + \mu(B\backslash A)$$

If $\mu(B\backslash A)$ is finite, the conclusion holds. If $\mu(B\backslash A) = \infty$, from equations (1.1.1) and (1.1.2), we get $\mu(A \cup B) = \mu(B) = \infty$, and the conclusion is again verified.

(iii) Since $A \subset B$, we have $B = A \cup (B\backslash A)$. However, A and $B\backslash A$ are disjoint, so $\mu(B) = \mu(A) + \mu(B\backslash A)$. By assumption, $\mu(A) < \infty$, which gives us $\mu(B) - \mu(A) = \mu(B\backslash A)$.

(iv) From (iii) We have $\mu(B) = \mu(A) + \mu(B\backslash A)$, which implies $\mu(B) \geqslant \mu(A)$.

(v) We define $A_0 := \varnothing$ and $B_k := A_k\backslash A_{k-1}$, where $k \in \mathbb{N}$. By assumption, (B_k) is a disjoint sequence in $\mathcal{A}$, and we have $\bigcup_{k=1}^{\infty} B_k = \bigcup_{j=1}^{\infty} A_j$ and $\bigcup_{k=1}^{m} B_k = A_m$. The σ-additivity of μ means that

$$\mu(\bigcup_j A_j) = \mu(\bigcup_k B_k) = \lim_{m\to\infty} \sum_{k=1}^{m} \mu(B_k) = \lim_{m\to\infty} \mu(\bigcup_{k=1}^{m} B_k) = \lim_{m\to\infty} \mu(A_m).$$

(vi) If(A_k) is a decreasing sequence of sets in $\mathcal{A}$, then $(A_1 \backslash A_k)$ is an increasing sequence. Furthermore, we obtain

$$A_1 \backslash (\bigcap_k A_k) \ = A_1 \cap \ (\bigcap_k A_k)^c = \bigcup_k (A_1 \cap A_k^c) \ = \bigcup_k (A_1 \backslash A_k)$$

From (iii) and (v), we get

$$\begin{aligned} \mu(A_1) - \mu(\bigcap_k A_k) = \mu(A_1 \backslash (\bigcap_k A_k)) \ &= \mu(\bigcup_k (A_1 \backslash A_k)) \\ = \lim_{m\to\infty} \mu(A_1 \backslash A_m) \ &= \mu(A_1) - \lim_{m\to\infty} \mu(A_m) \ , \end{aligned}$$

thus the conclusion holds.

(vii) Let $B_1 := A_1$ and $B_k := A_k \backslash (\bigcup_{j=1}^{k-1} A_j)$ for $k \geqslant 2$. The sequence (B_k) in $\mathcal{A}$ satisfies $\bigcup_k B_k = \bigcup_k A_k$ and $B_k \subset A_k$ for $k \in \mathbb{N}$. From (iv) and the σ-additivity of μ, we have

$$\mu(\bigcup_k A_k) \ = \mu(\bigcup_k B_k) \ = \sum_k \mu(B_k) \ \leqslant \sum_k \mu(A_k).$$

□

Remark 1.1.18. In Proposition 1.1.17(vi), the condition $\mu(A_1) < \infty$ cannot be omitted. Here is a counterexample: Consider $\Omega = \mathbb{N}$ and $A_n := \{m \in \mathbb{N} : m \geqslant n\}$. Let μ be the counting measure on $\mathbb{N}$, i.e., $\mu(A) := \#(A)$. Then $\mu(A_n) = \infty$ for every $n \in \mathbb{N}$ and $\bigcap_{n=1} A_n = \varnothing$. However, $\mu(A_n) \downarrow \mu(\varnothing) = 0$ does not hold.

Let $(\Omega, \ \mathcal{A}, \ \mu)$ be a measure space. A set $N \subset \Omega$ is called a null set (or more precisely, a μ-null set) if $N \in \mathcal{A}$ and $\mu(N) = 0$. We denote by $\mathcal{N}$ the collection of all null sets. From the monotonicity of the measure and theσ-subadditivity, we can easily deduce the following propositions:

$$\varnothing \in \mathcal{N};$$

$$A \in \mathcal{N}, \ A' \in \mathcal{A}, \ A' \subset A \ \Rightarrow \ A' \in \mathcal{N};$$

$$A_1, \ A_2, \ \ldots \ \in \mathcal{N} \ \Rightarrow \ \bigcup_{n\geqslant 1} A_n \in \mathcal{N}.$$

Due to the constraints of measurability, a subset of a null set is not necessarily a null set. This leads to the following concept of completeness.

Definition 1.1.19. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. Define $\mathcal{N}_\mu := \{A \in \mathcal{A} \ : \ \mu(A) = 0\}$, the collection of all μ-null sets. We say that $(\Omega, \mathcal{A}, \mu)$ is complete if $N \in \mathcal{N}_\mu$ and $M \subset N$ implies that $M \in \mathcal{A}$.

Exercises

1. Prove the Proposition1.1.8.

2. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, and let $f : \Omega \to \mathbb{R}$ be measurable. For $B \in \mathcal{B}(\mathbb{R})$, define $\nu(B) := \mu(f^{-1}(B))$. The measure ν is called the pushforward measure of μ under the map f.
(a) Prove that $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \nu)$ is a measure space.
(b) Let $\Omega = \mathbb{R}$, $\mathcal{A} = \mathcal{B}(\mathbb{R})$, $\mu = \lambda$, and $f(x) = x^3 - 1$. Compute $\nu([-1, 7])$.

3. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space.
Define $\mathcal{A}^*$: $=\{E \subset \Omega$: $\exists A, B \in \mathcal{A}$ such that$A \subset E \subset B$ and $\mu(B \backslash A) = 0 \}$.
For $E \in \mathcal{A}^*$, define $\mu^*(E) := \mu(A)$. Prove the following statements: a) $\mathcal{A}^*$ is a σ-algebra containing $\mathcal{A}$; b) μ^* is well-defined; c) μ^* s a measure on $\mathcal{A}^*$, and $(\Omega, \mathcal{A}^*, \mu^*)$ is complete; d) μ^* is an extension of μ, that is, $\mu^*(E) := \mu(E)$, $\forall E \in \mathcal{A}$. We call $(\Omega, \mathcal{A}^*, \mu^*)$ the completion of $(\Omega, \mathcal{A}, \mu)$.

4. Let $(\Omega, \mathcal{A})$ be a measurable space, and let $\mu : \mathcal{A} \to [0, \infty]$ be a mapping that satisfies the following conditions: $\mu(\varnothing) = 0$ and $\mu(A \cup B) = \mu(A) + \mu(B)$ for any disjoint sets $A, B \in \mathcal{A}$, and for any increasing sequence $(A_n) \subset \mathcal{A}$, we have $\lim_{n\to\infty} \mu(A_n) = \mu(A)$, where $A := \bigcup_{n\in\mathbb{N}} A_n$. Prove that μ is a measure.

1.2 Measurable Maps

Let f be a map from the space S to T. We define the preimage map f^{-1}: $2^T \to 2^S$ as follows:

$$f^{-1}B = \{s \in S; f(s) \in B\},\ B \subset T.$$

Note that f^{-1} preserves all set operations, that is, for any subsets B and B_k of T, we have

$$f^{-1}B^c = (f^{-1}B)^c, f^{-1}\bigcup_k B_k = \bigcup_k f^{-1}B_k, f^{-1}\bigcap_k B_k = \bigcap_k f^{-1}B_k.$$

For convenience, we introduce the notation

$$f^{-1}\mathcal{C} = \{f^{-1}B; B \in \mathcal{C}\},\ \mathcal{C} \subset 2^T.$$

Given two measurable spaces $(S,\ \mathcal{S})$ and $(T,\ \mathcal{T})$, we say that the map $f : S \to T$ is $\mathcal{S}/\mathcal{T}$-measurable or simply measurable, if $f^{-1}\mathcal{T} \subset \mathcal{S}$ (i.e.,$f^{-1}B \in \mathcal{S}$ for every $B \in \mathcal{T}$). The following lemma provides a necessary and sufficient condition for measurability.

Lemma 1.2.1. Let $(S,\ \mathcal{S})$ and $(T,\ \mathcal{T})$ be two measurable spaces, and let $\mathcal{C} \subset 2^T$ such that $\sigma(\mathcal{C}) = \mathcal{T}$. Then f is $\mathcal{S}/\mathcal{T}$-measurable if and only if $f^{-1}\mathcal{C} \subset \mathcal{S}$.

Proof. Let $\mathcal{T}' := \{B \subset T;\ f^{-1}B \in \mathcal{S}\}$. By the assumption, $\mathcal{C} \subset \mathcal{T}'$ and by Exercise 1(ii) of this section, $\mathcal{T}'$ 为 σ-algebra. Therefore,

$$\mathcal{T}' = \sigma(\mathcal{T}') \supset \sigma(\mathcal{C}) = \mathcal{T},$$

which verifies that $f^{-1}B \in \mathcal{S}$, $\forall B \in \mathcal{T}$. □

Lemma 1.2.2. Let $(S,\ \mathcal{S})$, $(T,\ \mathcal{T})$ and $(U,\mathcal{U})$ be measurable spaces, and let the maps $f : S \to T$ and $g : T \to U$ be measurable. Then the composite map $h = g \circ f : S \to U$is also measurable.

Proof. Let $C \in \mathcal{U}$. By the measurability of g, we have $B := g^{-1}C \in \mathcal{T}$. Noting that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ and the measurability of f, we have

$$h^{-1}C = (g \circ f)^{-1}C = f^{-1}\left(g^{-1}C\right) = f^{-1}B \in \mathcal{S}.$$

□

Exercises

1. Let $(\Omega, \mathcal{A})$ and $(\Omega', \mathcal{A}')$ be measurable spaces, $f : \Omega \to \Omega'$ be measurable, and μ be a measure on Ω with $\nu = \mu \circ f^{-1}$. Prove that f is measurable with respect to $\mathcal{A}^*$ and $\mathcal{A}'^*$. Here, $\mathcal{A}^*$ and $\mathcal{A}'^*$ are the completions of $\mathcal{A}$ and $\mathcal{A}'$, respectively.

1.3 Measurable Functions

Let $(\Omega, \mathcal{A})$ be a measurable space. In this section, we will introduce the concept of measurable functions on $(\Omega, \mathcal{A})$. In real analysis, the functions we consider are allowed to take the values +∞ and -∞, which can be convenient when discussing the limits of function sequences.

Definition 1.3.1. The extended real number set $\bar{\mathbb{R}} := [-\infty, +\infty]$. . The following rules are applied regarding the operations with $\pm\infty$:
1) For $\forall x \in \mathbb{R}$,

$$\begin{aligned} &x + \infty = \infty = \infty + x \\ &x - \infty = -\infty = -\infty + x \\ &+\infty + \infty = +\infty \\ &-\infty + (-\infty) = -\infty; \end{aligned}$$

2) For $\forall x > 0$,

$$\begin{aligned} &\pm x \cdot (+\infty) = \pm\infty = +\infty \cdot (\pm x) \\ &\pm x \cdot (-\infty) = \mp\infty = -\infty \cdot (\pm x) \\ &0 \cdot (\pm\infty) = (\pm\infty) \cdot 0 = 0 \\ &\frac{1}{\pm\infty} = 0 \end{aligned}$$

3) Operations that are meaningless: $\infty - \infty$, $\frac{\pm\infty}{\pm\infty}$, $-\infty - (-\infty)$.

Naturally, we need to establish a measurable structure on $\bar{\mathbb{R}}$.

Definition 1.3.2. $\mathcal{B}(\bar{\mathbb{R}}) := \sigma\left\{\mathcal{B}(\mathbb{R}) \cup \{+\infty\} \cup \{-\infty\}\right\}$.

Remark 1.3.3. If $B \subset \bar{\mathbb{R}}$, then $B^* \in \mathcal{B}(\bar{\mathbb{R}})$ if and only if $B^* = B \cup S$, where $B \in \mathcal{B}(\mathbb{R})$ and $S \in \{\varnothing, \{+\infty\}, \{-\infty\}, \{+\infty, -\infty\}\}$.

Definition 1.3.4. If a generalized real-valued function$f : \Omega \to \overline{\mathbb{R}}$ on $(\Omega, \mathcal{A})$ is measurable with respect to $\mathcal{A}/\mathcal{B}(\overline{\mathbb{R}})$, then we call it a measurable function on $(\Omega, \mathcal{A})$.

Notation: Suppose $f : \Omega \to \overline{\mathbb{R}}$ and $c \in \mathbb{R}$. Define $\{f > c\} := \{x \in \Omega : f(x) > c\} = f^{-1}((c, \infty))$, and similarly define the sets $\{f \geqslant c\}, \{f < c\}$ and $\{f \leqslant c\}$.

Theorem 1.3.5. *Let $(\Omega, \mathcal{A})$ be a measurable space. A function $f : \Omega \to \overline{\mathbb{R}}$ is said to be $\mathcal{A}$-measurable if one of the following conditions holds:*

1) $\{f > c\} \in \mathcal{A}$ $\quad \forall c \in \mathbb{R}$;

2) $\{f \geqslant c\} \in \mathcal{A}$ $\quad \forall c \in \mathbb{R}$;

3) $\{f \leqslant c\} \in \mathcal{A}$ $\quad \forall c \in \mathbb{R}$;

4) $\{f < c\} \in \mathcal{A}$ $\quad \forall c \in \mathbb{R}$.

Proof. Conditions 1) to 4) are mutually equivalent, because $\{f \geqslant c\} = \bigcap_{k=1}^{\infty} \{f > c - \frac{1}{k}\}$ $\{f < c\} = \{f \geqslant c\}^c \{f \leqslant c\} = \bigcap_{k=1}^{\infty} \{f < c + \frac{1}{k}\} \{f > c\} = \{f \leqslant c\}$, Therefore, 1)$\Rightarrow 2) \Rightarrow 3) \Rightarrow 4) \Rightarrow 1)$. Thus, we only need to prove that the measurability of f is equivalent to condition 1).

Since $\{f > c\} = f^{-1}((c, +\infty))$, according to Lemma 1.2.1, we only need to verify that $\mathcal{B}(\overline{\mathbb{R}})$ can be generated by $\Sigma := \{(c, +\infty] : c \in R\}$. Notice that (c,+∞] = (c,+∞)$\cup\{\infty\} \in \mathcal{B}(\overline{\mathbb{R}})$. Therefore, $\Sigma \subset \mathcal{B}(\overline{\mathbb{R}})$ and $\sigma(\Sigma) \subset \mathcal{B}(\overline{\mathbb{R}})$. We have $\{\infty\} = (\bigcup_{n=1}^{\infty}[-n, n])^c = \bigcap_{n=1}^{\infty}([-n, n]^c)$. Furthermore, for $a, b \in \mathbb{R}$ with $a < b$, we have$(a, b] = (a, +\infty] \backslash (b, +\infty] \in \sigma(\Sigma)$. So $\mathcal{B}(\overline{R}) \subset \sigma(\Sigma)$. Therefore, $\mathcal{B}(\overline{\mathbb{R}}) \subset \sigma(\Sigma)$ Hence, $\mathcal{B}(\overline{\mathbb{R}}) = \sigma(\Sigma)$. □

Example 1.3.6. (a) Suppose f is real-valued and constant. Then the set $\{x : f(x) > a\}$ can only be the empty set orΩ, so f is measurable.

(b) Let $B \subset \Omega$ be an arbitrary subset. The indicator function of$B$, $\mathbb{1}_B : \Omega \to \mathbb{R}$, is defined as follows:

$$\mathbb{1}_B(x) = \begin{cases} 1, & \text{if } x \in B; \\ 0, & \text{if } x \notin B. \end{cases}$$

Then the set $\{x : \mathbb{1}_B(x) > a\}$ can only be $\varnothing, B$ or Ω. Therefore, $\mathbb{1}_B(x)$ is measurable if and only if $B \in \mathcal{A}$.

(c) Suppose $\Omega = \mathbb{R}$, $\mathcal{A}$is the Borel σ-algebra and $f(x) = x$. Then $\{x : f(x) > a\} = (a, \infty)$, so f is measurable.

Proposition 1.3.7. *Consider measurable functions f, g on $(\Omega, \mathcal{A})$. Then the sets $\{f > g\}$, $\{f \geqslant g\}$, $\{f = g\}$ and $\{f \neq g\}$ all belong to $\mathcal{A}$.*

Proof. Note that $f(x) > g(x)$ holds if and only if there exists a rational number r such that $f(x) > r > g(x)$. Since the sets $\{f > r\}$ and $\{r > g\}$ belong to $\mathcal{A}$, we obtain

$$\{f > g\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{r > g\}) \in \mathcal{A}.$$

Furthermore,

$$\begin{aligned} &\{f \geqslant g\} = \{g > f\}^c \in \mathcal{A}, \\ &\{f = g\} = \{f \geqslant g\} \backslash \{f > g\} \in \mathcal{A}, \\ &\{f \neq g\} = \{f = g\}^c \in \mathcal{A}. \end{aligned}$$

□

Definition 1.3.8. For two functions $f, g : \Omega \to \bar{\mathbb{R}}$, define $\sup(f,g)$, $\inf(f,g) : \Omega \to \overline{\mathbb{R}}$ as follows:

$$\sup(f,g)(x) := \max\{f(x), g(x)\}, \qquad \sup(f,g)(x) := \min\{f(x), g(x)\}$$

for all $x \in \Omega$. Now, consider a sequence of functions $f_k : \Omega \to \bar{\mathbb{R}}$, $k \geqslant 1$. Define $\sup_k f_k$, $\inf_k f_k$: $\Omega \to \overline{\mathbb{R}}$ as follows:

$$(\sup_k f_k)(x) := \sup f_k(x),\ k \geqslant 1, \quad (\inf_k f_k)(x) := \inf f_k(x),\ k \geqslant 1.$$

Additionally, define $\limsup_k f_k, \liminf_k f_k : \Omega \to \overline{\mathbb{R}}$ as follows:

$$\limsup_k f_k := \inf_m(\sup_{k \geqslant m} f_k), \qquad \liminf_k f_k := \sup_m(\inf_{k \geqslant m} f_k).$$

The positive and negative parts of a function $f : \Omega \to \bar{\mathbb{R}}$ are defined as follows:

$$f^+ := \sup(f, 0), \quad f^- := \sup(-f, 0).$$

We have $f = f^+ - f^-$.

Theorem 1.3.9. *Let $f, g : \Omega \to \overline{\mathbb{R}}$ and $f_{\mathrm{k}} : \Omega \to \overline{\mathbb{R}}$, $k \geqslant 1$ be measurable functions. Then the following functions are also measurable:*

$$\sup(f,g),\ \inf(f,g),\ \sup_k f_k, \inf_k f_{\mathrm{k}},\ \limsup_k f_k,\ \liminf_k f_k.$$

Proof. The conclusion of the theorem can be derived from the following relations:

$$\{\sup_k f_k \leqslant c\} = \bigcap_{k \geqslant 1}\{f_k \leqslant c\} \quad \text{and} \quad \{\inf_k f_k \geqslant c\} = \bigcap_k \{f_k \geqslant c\}.$$

□

Definition 1.3.10. a) If f is a measurable function on $(\Omega, \mathcal{A})$ and $f(\omega) \in \mathbb{R}$, $\forall\ \omega \in \Omega$, then we call f real-valued measurable function.

b) We say that a function g on $(\Omega,\ \mathcal{A})$ is a simple function if g has a representation of the following form:

$$g(x) = \sum_{j=1}^{m} y_j \cdot 1_{B_j}(x),$$

where $m \in \mathbb{N}, y_j \in \mathbb{R}, B_j \in \mathcal{A}$, $\forall\ j = 1, \ldots, m$. If

$$g(x) = \sum_{k=0}^{n} z_k \cdot 1_{A_k}(x), \tag{1.3.1}$$

where $z_k \in \mathbb{R}, A_k \in \mathcal{A}$, $k = 0, \ldots, n$, and $\uplus_{k=0}^{n} A_k = \Omega$, then we call (1.3.1) a standard representation. Let $\varepsilon(\mathcal{A})$ and $\varepsilon^+(\mathcal{A})$ denote the set of all simple functions and non-negative simple functions on $(\Omega,\ \mathcal{A})$, respectively.

Example 1.3.11. (i). If h is a measurable function on $(\Omega,\ \mathcal{A})$ and $h(\Omega) = \{y_1, \cdots, y_m\}$, $m \in \mathbb{N}, y_j \in \mathbb{R},\ j = 1, \ldots, m$, then h is a simple function. Furthermore,

$$h(x) = \sum_{j=1}^{m} y_j \cdot \mathbb{1}_{A_j}(x), \quad \omega \in \Omega, \tag{1.3.2}$$

where

$$A_j\ := \{h = y_j\} = \{h \geqslant y_j\} \backslash \{h > y_j\} \in \mathcal{A}$$

and the sets A_j are pairwise disjoint (because the y_j are distinct), which means that (1.3.2) is a standard representation. It follows that g is a simple function if and only if g is a real-valued measurable function that takes only finitely many possible values.

(ii). $f, g \in \varepsilon(\mathcal{A}) \Rightarrow f \pm g, f \cdot g \in \varepsilon(\mathcal{A})$. The proof is as follows: Let

$$f = \sum_{j=0}^{m} y_j\ .\ 1_{A_j}\ g = \sum_{k=0}^{n} z_k\ .\ 1_{B_k}$$

be the standard representations f and g, respectively. Then:

$$f \pm g\ =\ \sum_{j=0}^{m} \sum_{k=0}^{n} (y_j \pm z_k)\ .\ 1_{A_j \cap B_k}$$

$$f \cdot g\ =\ \sum_{j=0}^{m} \sum_{k=0}^{n} y_j \cdot z_k \cdot 1_{A_j \cap B_k}$$

Note that $(A_j \cap B_k)_{j,k}$ is a partition of Ω, and: $f|_{A_j \cap B_k} = y_j$, if$A_j \cap B_k \neq \varnothing$; $g|_{A_j \cap B_k} = z_k$, if $A_j \cap B_k \neq \varnothing$.

(iii). $f \in \varepsilon(A) \Rightarrow f^+, f^-, |f| \in \varepsilon(\mathcal{A})$

Lemma 1.3.12. *(Sombrero) (a) Let $(\Omega,\ \mathcal{A})$ be a measurable space. Then, for any $\mathcal{A}/\mathcal{B}(\overline{\mathbb{R}})$-measurable function$u : \Omega \to \overline{\mathbb{R}},\ u \geqslant 0$, there exists a sequence $(f_j)_{j\in\mathbb{N}} \subset \varepsilon^+(\mathcal{A})$ such that $0 \leqslant f_1 \leqslant \ldots \leqslant f_n \leqslant \ldots \leqslant u$ and $u = \sup_{j\in\mathbb{N}} f_j$.*

(b) For any $\mathcal{A}/B(\overline{\mathbb{R}})$-measurable function$u : \Omega \to \overline{\mathbb{R}}$, there exists a simple function f_n such that $|f_n| \leqslant |f|$ and $f(\omega) = \lim_{n\to\infty} f_n(\omega)$, $\omega \in \Omega$.

Proof. (a) For $j \in \mathbb{N}$, define:

$$A_k^j := \begin{cases} \{k \cdot 2^{-j}\ \leqslant u <\ (k+1) \cdot 2^{-j}\} & k = 0,\ \ldots,\ j \cdot 2^j - 1 \\ \{u \geqslant j\} & k = j \cdot 2^j \end{cases}$$

Then:

(i). $f_j(x) := \sum_{k=0}^{j2^j} k \cdot 2^{-j} \cdot 1_{A_k^j}(x) \leqslant u(x)$

(ii). $|f_j(x) - u(x)| \leqslant 2^{-j}$, $\forall\ k \neq j \cdot 2^j (\Leftrightarrow x \notin \{u \geqslant j\})$

(iii). From

$$\{k \cdot 2^{-j}\ \leqslant u <\ (k+1) \cdot 2^{-j}\} = \{k \cdot 2^{-j}\ \leqslant u\} \cap \{u <\ (k+1) \cdot 2^{-j}\} \in \mathcal{A}$$

and $\{u \geqslant j\} \in \mathcal{A}$, it follows that $A_j^k \in \mathcal{A}$

(iv). $0 \leqslant f_j \leqslant f_{j+1} \leqslant \ldots \leqslant u, f_j \uparrow u$

(b) Let g_n and h_n be non-negative simple functions such that $g_n \uparrow f^+$, $h_n \uparrow f^-$. We can take $f_n = g_n - h_n$. □

Theorem 1.3.13. *If h_1 and h_2 are measurable functions on$(\Omega, \mathcal{A})$ then the following functions are also measurable: $h_1 + h_2$, $h_1 - h_2$, $h_1 h_2$, and h_1/h_2. Here, we assume that all the operations are well-defined, meaning that for any $\omega \in \Omega$, $h_1(\omega) + h_2(\omega)$ does not result in the indeterminate form $\infty - \infty$, and $h_1(\omega)/h_2(\omega)$ is not in the form ∞/∞ or $a/0$.*

Proof. Let s_{1n} , s_{2n} be simple functions such that $s_{1n} \to h_1, s_{2n} \to h_2$. Then $s_{1n} + s_{2n} \to h_1 + h_2$,

$$s_{1n} s_{2n} \mathbb{1}_{\{h_1 \neq 0\}} \mathbb{1}_{\{h_2 \neq 0\}} \to h_1 h_2$$

and

$$\frac{s_{1n}}{s_{2n} + (1/n)\mathbb{1}_{\{s_{2n}=0\}}} \to \frac{h_1}{h_2}.$$

Because

$$s_{1n} \pm s_{2n},$$

$$s_{1n} s_{2n} \mathbb{1}_{\{h_1 \neq 0\}} \mathbb{1}_{\{h_2 \neq 0\}},\ s_{1n}(s_{2n} + \frac{1}{n}\mathbb{1}_{\{s_{2n}=0\}})^{-1}$$

are all simple functions, and in particular, they are all measurable. Now that the target functions have all been expressed as limits of measurable functions, the conclusion follows from Theorem1.3.9. □

Exercises

1. Prove that f is a real-valued measurable function on $(\Omega,\ \mathcal{A})$ if and only if $f : (\Omega, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a measurable mapping.

2. Let $(\Omega, \mathcal{A})$ be a measurable space and $f_n : \Omega \to \overline{\mathbb{R}}$ be a sequence of measurable functions. Prove that the following sets M_i, $i = 1, 2, 3$, are all measurable, i.e., $M_i \in \mathcal{A}$:

$$M_1 := \{x \in \Omega : \lim_{n\to\infty} f_n(x) \text{ exists in } \overline{\mathbb{R}}, \text{ i.e. } \exists y \in \overline{\mathbb{R}} \text{ such that } f_n(x) \to y \text{ as } n \to \infty\},$$

$$M_2 := \{x \in \Omega : \lim_{n\to\infty} f_n(x) \text{ exists in } \mathbb{R}, \text{ i.e. } \exists y \in \mathbb{R} \text{ such that } f_n(x) \to y \text{ as } n \to \infty\},$$

$$M_3 := \{x \in \Omega : \lim_{n\to\infty} f_n(x) = 27\}.$$

3. Please check the correct statements below. Here, $(\Omega, \mathcal{A})$ is a measurable space and $f, g : \Omega \to \bar{\mathbb{R}}$ are functions.
   □ Suppose $\mathcal{A} := \{\varnothing, \Omega\}$. Then all measurable functions on $(\Omega, \mathcal{A})$ are constant functions.
   □ If $f + g$ is measurable, then f and g are both measurable.
   □ If f and g are measurable, then $\{\omega \in \Omega\ :\ f^2(\omega) \leqslant g(\omega)\} \in \mathcal{A}$.
   □ If f^2 is measurable, then f is also measurable.
   □ If f^3 is measurable, then f is also measurable.
   □ If f is measurable, then $\{f(B)\ :\ B \in \mathcal{A}\}$ is a σ-algebra on $\bar{\mathbb{R}}$.
   □ If f and g are measurable, then $f\mathbb{1}_{\{g>c\}}$ is also measurable, where $c > 0$ is a constant.

☐ Suppose that for every $n \in \mathbb{N}$, $f\mathbb{1}_{\{|f| \leqslant n\}}$ is measurable. Then f is also measurable.
☐ Suppose that for every $\alpha \in \mathbb{R}$, $f_\alpha$is measurable. Then $\sup_{\alpha \in \mathbb{R}} f_\alpha$ is also measurable.

1.4 Definition of the Integral

Let us first consider the integral of non-negative simple functions.

• Let $(\Omega,\ \mathcal{A},\ \mu)$ be a measure space.

• Let $\mathcal{M}(\mathcal{A})$ ($\mathcal{M}^+(\mathcal{A})$) denote the set of all (non-negative) measurable functions on$(\Omega,\ \mathcal{A})$, and let $\varepsilon(\mathcal{A})(\varepsilon^+(\mathcal{A}))$ denote the set of all (non-negative) simple functions on $(\Omega,\ \mathcal{A})$.

• Basic idea: For a simple function$f \geqslant 0$, the integral $\int f$ can intuitively be viewed as the volume under the graph of$f$, That is, if

$$f(x) = \sum_{j=0}^{m} y_j \ . \ 1_{A_j}\ (x)$$

and $\mathrm{U}_{j=0}^{m} A_j = \Omega, y_j \geqslant 0$, then

$$\int f = \sum_{j=0}^{m} y_j \cdot \mu(A_j).$$

A natural question arises: Is the expression for $\int f$ given here well-defined?

Lemma 1.4.1. *Suppose*

$$\sum_{j=0}^{m} y_j \ . \ 1_{A_j}\ \sum_{k=0}^{n} z_k \ . \ 1_{B_k}$$

are both standard representations of $f \in \varepsilon^+(\mathcal{A})$. Then:

$$\sum_{j=0}^{m} y_j \ . \ \mu(A_j) = \sum_{k=0}^{n} z_k \ . \ \mu(B_k)$$

Proof. By the definition of a standard representation, we have

$$\bigcup_{j=0}^{m} A_j = \Omega = \bigcup_{k=0}^{n} B_k$$

and

$$A_j = \bigcup_{k=0}^{n} (A_j \cap B_k)\ B_k = \bigcup_{j=0}^{m} (B_k \cap A_j)$$

Therefore

$$\begin{aligned} \sum_{j=0}^{m} y_j \ . \ \mu(A_j)\ &=\ \sum_{j=0}^{m} y_j \sum_{k=0}^{n} \mu(A_j \cap B_k) = \sum_{k=0}^{n} \sum_{j=0}^{m} y_j \ . \ \mu(A_j \cap B_k) \\ &= \sum_{k=0}^{n} \sum_{j=0}^{m} z_k \ . \ \mu(A_j \cap B_k) = \sum_{k=0}^{n} z_k \ . \ \mu(B_k) \end{aligned}$$

☐

Definition 1.4.2. Let $f = \sum_{j=0}^{m} y_j \cdot 1_{A_j} \in \varepsilon^+(\mathcal{A})$ be a standard representation. We define

$$I_\mu(f) := \sum_{j=0}^{m} y_j \cdot \mu(A_j) \in [0, \infty]$$

to be the integral of f.

We now define the integral of non-negative measurable functions.

Definition 1.4.3. If $u \in \mathcal{M}^+(\mathcal{A})$, then we define

$$\int u d\mu := \sup\{I_\mu(g); g \in \varepsilon^+(\mathcal{A}),\ 0 \leqslant g \leqslant u\}$$

Notation:

$$\int u d\mu = \int u(x)\mu(dx) = \int u(x) d\mu(x)$$

Remark 1.4.4. Due to the supremum, the right-hand side of the definition $\int u d\mu$ is a uniquely determined value (possibly ∞).

Lemma 1.4.5. $\int \cdot d\mu$ *is an extension of* I_μ *, that is, for all* $f \in \varepsilon^+(A)$*, we have:*

$$\int f d\mu = I_\mu(f)$$

Proof. Since $f \in \varepsilon^+(\mathcal{A})$ and $f \leqslant f$:

$$I_\mu(f) \leqslant \sup\{I_\mu(g); g \in \varepsilon^+(\mathcal{A}),\ g \leqslant f\} = \int f d\mu$$

On the other hand, if $g \in \varepsilon^+(\mathcal{A})$ and $g \leqslant f$, then $I_\mu(g) \leqslant I_\mu(f)$. Therefore:

$$\int f d\mu = \sup_{g \in \varepsilon^+(A), g \leqslant f} I_\mu(g) \leqslant I_\mu(f)$$

□

Definition 1.4.6. We define $\int_\Omega h d\mu = \int_\Omega h^+ d\mu - \int_\Omega h^- d\mu$, if the difference on the right-hand side is not of the form $+\infty - \infty$; if it is, we say that the integral of h does not exist. We say that a function h is μ-integrable (or simply integrable, if μ is clear from context) if and only if $\int_\Omega h d\mu$ is finite; equivalently, if and only if both $\int_\Omega h^+ d\mu$ and $\int_\Omega h^- d\mu$ are finite. We denote by $\mathcal{L}^1(\mu)$ the set of allμ-integrable functions. If $A \in \mathcal{F}$, we define

$$\int_A h d\mu = \int_\Omega h I_A d\mu.$$

Theorem 1.4.7. *(a) If* $\int_\Omega hd\mu$ *exists and* $c \in \mathbb{R}$*, then* $\int_\Omega chd\mu$ *also exists and equals* $c\int_\Omega hd\mu$.

(b) If $g(\omega) \geqslant h(\omega)$ *for all* ω*,then* $\int_\Omega gd\mu \geqslant \int_\Omega hd\mu$*, that is: If* $\int_\Omega hd\mu$ *exists and is greater than* $-\infty$ *, then* $\int_\Omega gd\mu$ *exists and* $\int_\Omega gd\mu \geqslant \int_\Omega hd\mu$*; If* $\int_\Omega gd\mu$ *exists and is less than* $+\infty$*, then* $\int_\Omega hd\mu$ *exists and* $\int_\Omega hd\mu \leqslant \int_\Omega gd\mu$*; If both integrals exist, then* $\int_\Omega gd\mu \geqslant \int_\Omega hd\mu$*, regardless of whether they are finite or infinite.*

(c) If $\int_\Omega hd\mu$ *exists, then* $|\int_\Omega hd\mu| \leqslant \int_\Omega |h|d\mu$.

(d) If $h \geqslant 0$ *and* $B \in \mathcal{F}$*, then* $\int_B hd\mu = \sup\{\int_B sd\mu : 0 \leqslant s \leqslant h, s$ *is a simple function}.*

(e) If $\int_\Omega hd\mu$ *exists, then* $\int_A hd\mu$ *exists for every* $A \in \mathcal{F}$*; if* $\int_\Omega hd\mu$ *is finite, then* $\int_A hd\mu$ *is finite for ever* $A \in \mathcal{F}$.

Proof. (a) It is clear that the proposition holds when h is a simple function. If h is non-negative and $c > 0$, then

$$\begin{aligned}\int_\Omega chd\mu &= \left\{\int_\Omega sd\mu; 0 \leqslant s \leqslant ch, s \text{ is a simple function}\right\} \\ &= c\sup\int_\Omega \frac{s}{c}d\mu; 0 \leqslant \frac{s}{c} \leqslant h, \frac{s}{c} \text{ is a simple function} = c\int_\Omega hd\mu.\end{aligned}$$

In general, if $h = h^+ - h^-$ and $c > 0$, then $(ch)^+ = ch^+, (ch)^- = ch^-$; Thus, from the result we just proved, we have $\int_\Omega chd\mu = c\int_\Omega h^+d\mu - c\int_\Omega h^-d\mu$, thus, $\int_\Omega chd\mu = c\int_\Omega hd\mu$. If $c < 0$, then $(ch)^+ = -ch^-$, $(ch)^- = -ch^+$, so

$$\int_\Omega chd\mu = -c\int_\Omega h^-d\mu + c\int_\Omega h^+ d\mu = c\int_\Omega hd\mu.$$

(b) If g and h are non-negative and $0 \leqslant s \leqslant h, s \text{ is a simple function}$, then $0 \leqslant s \leqslant g$; Thus, $\int_\Omega hd\mu \leqslant \int_\Omega gd\mu$. In general, $h \leqslant g$ means that $h^+ \leqslant g^+, h^- \geqslant g^-$. If $\int_\Omega hd\mu > -\infty$, we have $\int_\Omega g^-d\mu \leqslant \int_\Omega h^-d\mu < \infty$; Thus, $\int_\Omega gd\mu$ exists and equals

$$\int_\Omega g^+d\mu - \int_\Omega g^-d\mu \geqslant \int_\Omega h^+d\mu - \int_\Omega h^-d\mu = \int_\Omega hd\mu.$$

The proof in the case where $\int_\Omega gd\mu < \infty$ is similar, so we omit it.

(c) Since $-|h| \leqslant h \leqslant |h|$, by (a) and (b), we have $-\int_\Omega |h|d\mu \leqslant \int_\Omega hd\mu \leqslant \int_\Omega |h|d\mu$, so the proposition holds. (Note that $|h|$ is measurable, because $h = h^+ + h^-$.)

(d) If $0 \leqslant s \leqslant h$, then by (b), we have $\int_B sd\mu \leqslant \int_B hd\mu$; Thus,

$$\int_B hd\mu \geqslant \sup\{\int_B sd\mu : 0 \leqslant s \leqslant h\}.$$

If $0 \leqslant t \leqslant hI_B$, and t is a simple function, then $t = tI_B \leqslant h$, so

$$\int_\Omega t d\mu \leqslant \sup\left\{\int_\Omega sI_B d\mu : \ 0 \leqslant s \leqslant h, \, s \text{ is a simple function}\right\}.$$

Taking the supremum over all t, we get

$$\int_B h d\mu \leqslant \sup\{\int_B s d\mu : \ 0 \leqslant s \leqslant h, s \text{ is a simple function}\}.$$

(e) This result can be derived from (b) and $(hI_A)^+ = h^+ I_A \leqslant h^+$, $(hl_A)^- = h^- I_A \leqslant h^-$. □

1.5 Properties of the Integral

In this section, we further investigate the properties of the integral. The results in this section will be used repeatedly in the future. As in the previous section, $(\Omega, \mathcal{F}, \mu)$ is an arbitrary measure space. Unless otherwise stated, the functions considered in this section are generalized real-valued functions, i.e., mappings from Ω to $\overline{\mathbb{R}}$.

Theorem 1.5.1. *Let f_1, f_2 , . . . be a sequence of non-negative measurable and monotonically increasing functions. Define $f(\omega) = \lim_{n\to\infty} f_n(\omega)$, $\omega \in \Omega$. Then, $\int_\Omega f_n d\mu \uparrow \int_\Omega f d\mu$. (This theorem states that if $0 \leqslant f_n \uparrow f$, then $\int_\Omega f_n d\mu \uparrow \int_\Omega f d\mu$.)*

Proof. By the monotonicity of the integral, $\int_\Omega f_n d\mu$ is monotonically increasing with respect to n, so $\lim_{n\to\infty} \int f_n \mathrm{d}\mu \leqslant \int f \mathrm{d}\mu$.

Let $h \geqslant 0$ be a simple function with a standard representation $h = \sum_{i=1}^m x_i \mathbb{1}_{A_i}$ and $h \leqslant f$. Let $\varepsilon > 0$. For the simple function

$$h_n := (h - \varepsilon)^+ \cdot \mathbb{1}_{\{f_n > f - \varepsilon\}},$$

we have $0 \leqslant h_n \leqslant f_n$ (If $h_n(\omega) > f_n(\omega)$, then $h(\omega) - \varepsilon > f_n(\omega)$ and $f_n(\omega) > f(\omega) - \varepsilon$, leading to $f(\omega < f_n(\omega) + \varepsilon < h(\omega)$, a contradiction). By the definition of the integral, we get

$$\sum_{i=1}^m (x_i - \varepsilon)^+ \mu\left(A_i \cap \{f_n > f - \varepsilon\}\right) = \int h_n \mathrm{d}\mu \leqslant \int f_n \mathrm{d}\mu.$$

By the assumption $\{f_n > f - \varepsilon\} \uparrow \Omega$, we have $\mu\left(A_i \cap \{f_n > f - \varepsilon\}\right) \to \mu\left(A_i\right)$, where we applied the continuity of measure from below. Therefore,

$$\sum_{i=1}^m (x_i - \varepsilon)^+ \mu\left(A_i\right) \leqslant \lim_{n\to\infty} \int f_n \mathrm{d}\mu,$$

Now, letting $\varepsilon \to 0$, we obtain $\int h \mathrm{d}\mu = \sum_{i=1}^m x_i \mu\left(A_i\right) \leqslant \lim_{n\to\infty} \int f_n \mathrm{d}\mu$. By the definition of the integral and the arbitrariness of h, we conclude $\int f \mathrm{d}\mu \leqslant \lim_{n\to\infty} \int f_n \mathrm{d}\mu$. □

Theorem 1.5.2. *Let h be a measurable function and suppose that $\int_\Omega h d\mu$ exists. Define $v(B) = \int_B h d\mu, B \in \mathcal{F}$, Then, v is countably additive on $\mathcal{F}$; thus, if $h \geqslant 0, v$ is a measure on $\mathcal{F}$.*

Proof. Left as an exercise. □

Theorem 1.5.3. *Let f and g be measurable, and $f + g$ is everywhere defined. If $\int_\Omega f d\mu$, $\int_\Omega f d\mu$ exist and $\int_\Omega f d\mu + \int_\Omega g d\mu$ is well-defined (not $+\infty - \infty$ or $-\infty + \infty$), then*

$$\int_\Omega (f+g) d\mu = \int_\Omega f d\mu + \int_\Omega g d\mu.$$

In particular, if f and g are integrable, then $f + g$ is also integrable.

Proof. First, if f and g are non-negative simple functions, the conclusion can be directly derived from the definition of the integral.

Step 2: Let f and g be non-negative measurable functions. Then, there exist non-negative simple functions t_n and u_n such that $t_n \uparrow f$ and $u_n \uparrow g$. Thus $0 \leqslant s_n = t_n + u_n \uparrow f + g$. By the first step, $\int_\Omega s_n d\mu = \int_\Omega t_n d\mu + \int_\Omega u_n d\mu$; Applying the Monotone Convergence Theorem, we obtain $\int_\Omega (f+g) d\mu = \int_\Omega f d\mu + \int_\Omega g d\mu$.

Step 3: If $f \geqslant 0, g \leqslant 0, h = f + g \geqslant 0$ (so g must be finite), we have $f = h + (-g)$; hence, $\int_\Omega f d\mu = \int_\Omega h d\mu - \int_\Omega g d\mu$. If $\int_\Omega g d\mu$ is finite, then $\int_\Omega h d\mu = \int_\Omega f d\mu + \int_\Omega g d\mu$. If $\int_\Omega g d\mu = -\infty$, then since $h \geqslant 0$,

$$\int_\Omega f d\mu \geqslant -\int_\Omega g d\mu = \infty,$$

so $\int_\Omega f d\mu + \int_\Omega g d\mu$ is not well-defined, which contradicts the assumption. Similarly, if $f \geqslant 0, g \leqslant 0, h \leqslant 0$, we have$-g \geqslant 0, -f \leqslant 0, -h = -f - g \geqslant 0$, applying the result just proven,

$$\int_\Omega h d\mu = -\int_\Omega -h d\mu = -\left(\int_\Omega -f d\mu + \int_\Omega (-g)\, d\mu \right) = \int_\Omega f d\mu + \int_\Omega g d\mu.$$

Let

$$\begin{aligned}
E_1 &= \{\omega : f(\omega) \geqslant 0,\ g(\omega) \geqslant 0\}, \\
E_2 &= \{\omega : f(\omega) \geqslant 0,\ g(\omega) < 0,\ h(\omega) \geqslant 0\}, \\
E_3 &= \{\omega : f(\omega) \geqslant 0,\ g(\omega) < 0,\ h(\omega) < 0\}, \\
E_4 &= \{\omega : f(\omega) < 0,\ g(\omega) \geqslant 0,\ h(\omega) \geqslant 0\}, \\
E_5 &= \{\omega : f(\omega) < 0,\ g(\omega) \geqslant 0,\ h(\omega) < 0\}, \\
E_6 &= \{\omega : f(\omega) < 0,\ g(\omega) < 0\}.
\end{aligned}$$

From the previously proven result, we have

$$\int_{E_i} h d\mu = \int_{E_i} f d\mu + \int_{E_i} g d\mu.$$

According to Theorem1.5.2,

$$\int_\Omega f d\mu = \sum_{i=1}^{6} \int_{E_i} f d\mu, \int_\Omega g d\mu = \sum_{i=1}^{6} \int_{E_i} g d\mu,$$

so

$$\int_\Omega f d\mu + \int_\Omega g d\mu = \sum_{i=1}^{6} \int_{E_i} h d\mu.$$

If $\int_\Omega h d\mu$ exists, then by Theorem1.5.2, we obtain $\int_\Omega f d\mu + \int_\Omega g d\mu = \int_\Omega h d\mu$, Thus, the theorem is proven.

We only need to verify that $\int_\Omega h^+ d\mu$ and $\int_\Omega h^- d\mu$ cannot both be infinite. If this were not the case, that is, if

$$\int_{E_i} h^+ d\mu = \int_{E_j} h^- d\mu = \infty$$

for some i, j, then $\int_{E_i} h d\mu = \infty$, $\int_{E_j} h d\mu = -\infty$. Thus, either $\int_{E_i} f d\mu$ or $\int_{E_j} g d\mu = \infty$; Without loss of generality, assume

$$\int_\Omega f d\mu = \infty. \tag{1.5.1}$$

(Note that $\int_\Omega f^+ d\mu \geqslant \int_{E_i} f^+ d\mu$.). Similarly, we argue for index j to obtain $\int_\Omega g d\mu = -\infty$. Then, from (1.5.1), we get

$$\int_\Omega f d\mu + \int_\Omega g d\mu = \infty - \infty,$$

which leads to a contradiction. Therefore, $\int_\Omega h^+ d\mu$ and $\int_\Omega h^- d\mu$ must both be finite. □

Corollary 1.5.4. *(a) If h_1, h_2 , . . . are non-negative measurable functions, then*

$$\int_\Omega (\sum_{n=1}^{\infty} h_n) d\mu = \sum_{n=1}^{\infty} \int_\Omega h_n d\mu.$$

(b) A measurable function h is integrable if and only $|h|$ is integrable.

(c) If g and h are measurable functions and $|g| \leqslant h$, and $h$is integrable, then g is integrable.

Proof. (a) Since $\sum_{k=1}^{n} h_k \uparrow \sum_{k=1}^{\infty} h_k$, by the monotone convergence theorem, the result follows.

(b) Since $|h| = h^+ + h^-$, the result follows directly from the definition of the integral.

(c) By the monotonicity of the integral, $|g|$ is integrable. By (b), the result follows. □

Now we introduce the concept of "almost everywhere" (a.e.). As before, let $\mathcal{N}$ be the collection of sets of measure 0, i.e., the collection of measurable sets with measure zero. Let $\Pi(\omega)$ be a statement that depends on $\omega \in \Omega$. We say that Π holds almost everywhere if there exists a set $B \in \mathcal{N}$ such that $\Pi(\omega)$ holds for all $\omega \notin B$. For example, a measurable function f is almost everywhere finite if

$$\{\omega : \ f(\omega) = \infty \text{ or } -\infty\} \subset B \in \mathcal{N}.$$

Theorem 1.5.5. *Let f, g, h be measurable functions.*

(a) If $f = 0$ almost everywhere (a.e.), then $\int_\Omega f d\mu = 0$.

(b) If $g = h$ a.e. and $\int_\Omega g d\mu$ exists, then $\int_\Omega h d\mu$ also exists, and $\int_\Omega g d\mu = \int_\Omega h d\mu$.

Proof. (a) If $f = \sum_{i=1}^{n} x_i I_{A_j}$ is a simple function with standard representation, then $x_i \neq 0$ implies that $\mu(A_i) = 0$. Therefore $\int_\Omega f d\mu = 0$. If $f \geqslant 0$ and $0 \leqslant s \leqslant f$,, with s being a simple function, then $s = 0$ a.e., so $\int_\Omega s d\mu = 0$. Thus, $\int_\Omega f d\mu = 0$. If $f = f^+ - f^-$, then both f^+ and f^- are less than or equal to $|f|$ (=0 a.e.), so the result also holds.

(b) Let $\mathrm{A} = \{\omega : g(\omega) = h(\omega)\}, B = A^c$. Then, we have $g = gI_A + gI_B$, $h = hI_A + hI_B = gI_A + hI_B$. Since $gI_B = hI_B = 0$ outside B , and $B \in \mathcal{N}$, the conclusion holds. □

According to Theorem1.5.5, in a discussion about integrals, we can generally replace the condition that holds everywhere with the condition that holds almost everywhere, and the conclusion remains unchanged. For example, the following more general monotone convergence theorem holds: If $0 \leqslant h_n \uparrow h$ a.e., then $\int_\Omega h_n d\mu \uparrow \int_\Omega h d\mu$.

Another example: If g and h are measurable and $g \geqslant h$ a.e., then $\int_\Omega g d\mu \geqslant \int_\Omega h d\mu$, which means that the conclusion of Theorem1.4.7(b) still holds.

Theorem 1.5.6. *Let h be measurable.*

(a) If h is measurable, then h is finite almost everywhere.

$(\mathrm{b}\rangle$ If $h \geqslant 0$ and $\int_\Omega h d\mu = 0$, then $h = 0$ a.e.

Proof. (a) Let $A = \{\omega : \ |h(\omega)| \ = \infty\}$. If $\mu(A) > 0$, then $\int_\Omega |h| d\mu \geqslant \int_A |h| d\mu = \infty \cdot \mu(A) = \infty$, which leads to a contradiction.

(b) Let $B = \{\omega : h(\omega) > 0\}, B_n = \{\omega : h(\omega) \geqslant 1/n\} \uparrow B$. We have $0 \leqslant hI_{B_n} \leqslant hI_B = h$; by the monotonicity of the integral, $\int_{B_n} h d\mu = 0$. However, $\int_{B_n} h d\mu \geqslant (1/n)\mu(B_n)$, so $\mu(B_n) = 0$ for all n. Therefore, by the continuity of measure from below, we have $\mu(B) = 0$. □

The following result is called *Fatou's Lemma*.

Theorem 1.5.7. *Assume f_n are non-negative measurable functions. Then*

$$\int_\Omega \liminf_{n\to\infty} f_n \leqslant \liminf_{n\to\infty} \int_\Omega f_n.$$

Proof. Let $g_n = \inf_{i\geqslant n} f_i$. Then g is non-negative and $g_n$monotonically converges to $\liminf_n f_n$. Clearly, for$n \leqslant i$, we have $g_n \leqslant f_i$ and therefore $\int g_n \leqslant \int f_i$. Thus,

$$\int g_n \leqslant \inf_{i\geqslant n} \int f_i. \tag{1.5.2}$$

If we let $n \to \infty$ in equation(1.5.2), the left-hand side converges to $\int \liminf_n f_n$(by the Monotone Convergence Theorem), and the right-hand side converges to $\liminf_n \int f_n$. □

Fatou's Lemma is often used in this way: Assume that $f_n \to f$ and $\sup_n \int |f_n| \leqslant K < \infty$. Then $|f_n| \to |f|$, and by Fatou's Lemma, $\int |f| \leqslant K$.

We now introduce the most important result in integration theory, the Dominated Convergence Theorem.

Theorem 1.5.8. *Assume that$f, f_n$are measurable and $f_n \to f$ a.e. Assume there exists a non-negative integrable function g such that $|f_n| \leqslant g$ a.e. Then,*

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

Proof. Without loss of generality, assume that$f_n \to f$ and $|f_n| \leqslant g$holds almost everywhere. Since $f_n + g \geqslant 0$, by Fatou's Lemma, we have

$$\int f + \int g = \int (f+g) \leqslant \liminf_{n\to\infty} \int (f_n + g) = \liminf_{n\to\infty} \int f_n + \int g.$$

Since g is *integrable*, we obtain

$$\int f \leqslant \liminf_{n\to\infty} \int f_n. \tag{1.5.3}$$

Similarly, since $g - f_n \geqslant 0$, we have

$$\int g - \int f = \int (g-f) \leqslant \liminf_{n\to\infty} \int (g - f_n) = \int g + \liminf_{n\to\infty} \int (-f_n),$$

so

$$-\int f \leqslant \liminf_{n\to\infty} \int (-f_n) = -\limsup_{n\to\infty} \int f_n.$$

Thus,

$$\int f \geqslant \limsup_{n\to\infty} \int f_n,$$

Combining this with (1.5.3), the proof is complete. □

Exercise

1. Let $f, g : \Omega \to \bar{\mathbb{R}}$ be measurable functions, and $f + g$ is well-defined almost everywhere. If $\int_\Omega f d\mu$ and $\int_\Omega g d\mu$ exist, and $\int_\Omega f d\mu + \int_\Omega g d\mu$ is well-defined (i.e., not in the form $\infty - \infty$ or$-\infty + \infty$), then $\int_\Omega (f + g) d\mu$ must also exist.

2. (i) Let f_1, f_2, $\cdots$ be non-negative measurable functions. Prove
$$\int_\Omega (\sum_{n=1}^{\infty} f_n) d\mu = \sum_{n=1}^{\infty} \int_\Omega f_n d\mu.$$
(ii) Prove that $f : \Omega \to \bar{\mathbb{R}}$ is integrable if and only if $|f|$ is integrable.
(iii) Let f and g be measurable functions, and $|g| \leqslant f$. Prove that if f is integrable, then g is also integrable.

3. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. Prove that every μ-integrable function $f : \Omega \to \bar{\mathbb{R}}$ on Ω is μ-almost-everywhere-real-valued. Furthermore, the restriction of μ on $\{f \neq 0\}$ is a σ-finite measure.

4. Let f_1, f_2, $\cdots$ be measurable functions on $(\Omega, \mathcal{A}, \mu)$. Prove that if $\sum_{n=1}^{\infty} \int_\Omega |f_n| d\mu < \infty$, then $\sum_{n=1}^{\infty} f_n < \infty$ holds almost everywhere with respect to $[\mu]$and $\int_\Omega (\sum_{n=1}^{\infty} f_n) d\mu = \sum_{n=1}^{\infty} \int_\Omega f_n d\mu$.

5. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $\{A_n\}_{n\in\mathbb{N}}$ be a sequence of sets in $\mathcal{A}$. Define
$$\liminf_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m, \qquad \limsup_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m.$$
Prove that:
$$\mu(\liminf_{n\to\infty} A_n) \leqslant \liminf_{n\to\infty} \mu(A_n).$$
Ifμ is a finite measure, then also
$$\limsup_{n\to\infty} \mu(A_n) \leqslant \mu(\limsup_{n\to\infty} A_n).$$

6. Let μ be a σ-finite measure on the measurable space $(\Omega, \mathcal{A})$. If f and g are measurable, $\int_\Omega f d\mu$ and $\int_\Omega g d\mu$ exist, and $\int_A f d\mu \leqslant \int_A g d\mu$ for any $A \in \mathcal{A}$, then it follows that $f \leqslant g$ almost everywhere.

7. Let $f : \Omega \to \bar{\mathbb{R}}_+$ be integrable and $g : \Omega \to \bar{\mathbb{R}}$ be measurable. Define $\varphi_f(A) := \int_A f d\mu$, $A \in \mathcal{A}$. Prove that φ_f is a measure on $(\Omega, \mathcal{A})$. Furthermore, prove that g is φ_f-integrable if and only if gf is μ-integrable. In this case, we have
$$\int_\Omega g \, d\varphi_f = \int_\Omega fg \, d\mu.$$

8. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $f \in \mathcal{L}^1(\mu)$. Prove that for any $\varepsilon > 0$, there exists $A \in \mathcal{A}$ such that $\mu(A) < \infty$ and $|\int_A f d\mu - \int f d\mu| < \varepsilon$.

1.6 Dynkin's arguments

According to the usual assumption, let Ω be a non-empty set, and let $\mathcal{D}$ be a collection of subsets of Ω. We call $\mathcal{D}$ a Dynkin system when it satisfies the following three conditions.

(i) $\Omega \in \mathcal{D}$,

(ii) $A \in \mathcal{D} \Rightarrow A^{\mathrm{c}} \in \mathcal{D}$,

(iii) When $A_1, A_2, \ldots \in \mathcal{D}$ and are mutually disjoint, we have $\bigcup_{n \geqslant 1} A_n \in \mathcal{D}$

Remark 1.6.1. (1) Every σ-algebra is a Dynkin system.

(2) Let $\mathcal{D}$ be a Dynkin system. Similar to the property 1.1.6, we have $\varnothing \in \mathcal{D}$ and $\mathcal{D}$ is closed under finite unions of disjoint sets, i.e., if $A,\ B \in \mathcal{D}$ satisifies $A \cap B = \varnothing$, then $A \cup B \in \mathcal{D}$.

(3) Similar to the properties of σ-algebras, the intersection of Dynkin systems is still a Dynkin system. Similar to Theorem 1.1.9, for any collection of subsets $\mathcal{C}$, there exists the smallest Dynkin system containing $\mathcal{C}$, denoted by $\delta(\mathcal{C})$. When we want to prove that all elements in a σ-algebra satisfy a certain property, we use Dynkin systems as a tool to solve the problem.

Theorem 1.6.2. *Let $\mathcal{E}$ be a collection of subsets of Ω. Suppose that $\mathcal{E}$ is closed under finite intersections (i.e., if $A, B \in \mathcal{E}$, then $A \cap B \in \mathcal{E}$) and that $\mathcal{E} \subset \mathcal{D}$, where $\mathcal{D}$ is a Dynkin system. Then, we have $\sigma(\mathcal{E}) \subset \mathcal{D}$.*

Proof. Without loss of generality, we can assume that $\mathcal{D} = \delta(\mathcal{E})$. We first prove that $\mathcal{D}$ is closed under finite intersections. For each $D \in \mathcal{D}$, define

$$\mathcal{D}_D := \{A \in \mathcal{D} :\ A \cap D \in \mathcal{D}\}.$$

Then, $\mathcal{D}_D$ is a Dynkin system: Properties (i) and (iii) are clearly satisfied. If $A \in \mathcal{D}_D$, then for the union of two disjoint sets, $(A \cap D) \cup D^c$ belongs to $\mathcal{D}$, so its complement, $A^c \cap D$, also belongs to $\mathcal{D}$, satisfying condition (ii). Now consider $E \in \mathcal{E}$, then $\mathcal{E} \subset \mathcal{D}_E$ because $\mathcal{E}$ is closed under finite intersections. By the assumption that $\mathcal{D}$ is the smallest Dynkin system containing $\mathcal{E}$, we know that $\mathcal{D}_E = \mathcal{D}$. In other words, for any $D \in \mathcal{D}$ and $E \in \mathcal{E}$, we have $D \cap E \in \mathcal{D}$. Therefore, for all $D \in \mathcal{D}$, we have $\mathcal{E} \subset \mathcal{D}_D$. By the minimality of $\mathcal{D}$, we conclude that for all $D \in \mathcal{D}$, $\mathcal{D}_D = \mathcal{D}$. Thus, $\mathcal{D}$ is closed under finite intersections.

Assume that $A_1, A_2, \ldots$ all belong to $\mathcal{D}$. Note that we can write $\bigcup_{n \geqslant 1} A_n$ as the union of disjoint sets, because

$$\bigcup_{n \geqslant 1} A_n = A_1 \ \cup \bigcup_{n \geqslant 2} (A_n \cap A_1^c \cap \cdots \cap A_{n-1}^c) \ 。$$

Therefore, $\mathcal{D}$ is a σ-algebra. Since $\sigma(\mathcal{E})$ is the smallest σ-algebra containing $\mathcal{E}$, the assumption holds. □

This theory is often applied in the following situation: We usually want all sets in a σ-algebra $\mathcal{A}$ to satisfy some property Π. To achieve this, we typically proceed in two steps:
1) First, find a generating class $\mathcal{E}$ of $\mathcal{A}$ that satisfies finite intersections and prove that the sets in $\mathcal{E}$ satisfy property Π;
2) Consider the class of all sets $\mathcal{D}$ that satisfy property Π, and then prove that $\mathcal{D}$ is a Dynkin system. By applying 1.6.2, we obtain $\mathcal{A} \subset \mathcal{D}$, and all sets in $\mathcal{A}$ will possess property Π. This technique is known as Dynkin's argument.

Using Dynkin's argument, we can prove the following result regarding the uniqueness of the measure.

Theorem 1.6.3. *Let $(\Omega, \mathcal{A})$ be a measurable space, and let $\mathcal{E}$ be a generating class of $\mathcal{A}$ that satisfies the finite intersection property.*

a)Let μ and ν be finite measures on the measurable space$(\Omega,\ \mathcal{A})$, satisfying:

$$\mu(\Omega) = \nu(\Omega) \quad \textit{and} \quad \mu(E) = \nu(E)\,, \qquad \forall E \in \mathcal{E}.$$

then we have $\mu = \nu$.

b) Let μ and ν be measures onthe measurable space$(\Omega,\ \mathcal{A})$, satisfying that for any $E \in \mathcal{E}$, we have $\mu(E) = \nu(E)$. Assume that $(E_n)_{n\in\mathbb{N}} \subset \mathcal{E}$

And suppose that

$$\bigcup_{n=1}^{\infty} E_n = \Omega \quad \textit{and} \quad \mu(E_n) = \nu(E_n) < \infty\,, \qquad \forall n \in \mathbb{N}.$$

Then, we have$\mu = \nu$.

Proof. • Consider $j \in \mathbb{N}$. Define

$$\mathcal{D}_j \ := \{A \in \mathcal{A} : \mu(E_j \cap A) \ = \nu(E_j \cap A)\}$$

Then $\mathcal{D}_j$is a Dynkin system:

(i). (D1) Clearly holds.

(ii). (D2) Let $A \in \mathcal{D}_j$, then

$$\begin{aligned}\mu(A^c \cap E_j) \ &= \ \mu(E_j \backslash A) \ = \mu(E_j) - \mu(E_j \cap A) \qquad (< \infty)\\ &= \ \nu(E_j) - \nu(E_j \cap A) \ = \nu(E_j \cap A^c)\end{aligned}$$

so $A^c \in \mathcal{D}_j$.

(iii). (D3) Let$(A_k)_{k\in\mathbb{N}} \subset \mathcal{D}_j$ be a sequence of disjoint sets, then

$$\begin{aligned}\mu(\bigcup_{k\in\mathbb{N}} A_k \cap E_j) &= \ \sum_{k\in\mathbb{N}} \mu(A_k \cap E_j)\\ &= \ \sum_{k\in\mathbb{N}} \nu(A_k \cap E_j) = \nu(\bigcup_{k\in\mathbb{N}} A_k \cap E_j)\end{aligned}$$

which shows that $\bigcup_{k\in\mathbb{N}} A_k \in \mathcal{D}_j$.

• For any j, we have shown that $\mathcal{D}_j$ is a Dynkin system. Since $\mathcal{G}$ is closed under finite intersections, we have $\mathcal{D}_j \supset \mathcal{E}$ and

$$\mathcal{D}_j \supset \sigma(\mathcal{E}) = \mathcal{A}_\circ$$

From the definition, we have $\mathcal{A} \supset \mathcal{D}_j$, so for any j, $\mathcal{A} = \mathcal{D}_j$. In other words, for any $A \in \mathcal{A}$ and all $j \in \mathbb{N}$, we have

$$\mu(A \cap E_j) = \nu(A \cap E_j)$$

Therefore, for any $A \in \mathcal{A}$:

$$\mu(A) = \sup_{j\in\mathbb{N}} \mu(A \cap E_j) = \sup_{j\in\mathbb{N}} \nu(A \cap E_j) = \nu(A).$$

□

Exercises

1. Let $F : \mathbb{R} \to \mathbb{R}$ be the distribution function in the usual sense, i.e., F is non-decreasing and right-continuous. Prove that there exists, and at most one, measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that
$$\mu((a, b]) = F(b) - F(a), \quad \forall a < b,\ a, b \in \mathbb{R}_{\circ}$$
The existence of such a measure μ will be introduced in later studies, and we refer to it as the *Lebesgue-Stieltjes* measure corresponding to the distribution function F.

2. (Uniqueness of Measure) Suppose $\mathcal{E}$ is closed under $\bigcap$ and generates the σ-algebra $\mathcal{A}$ on Ω. Let μ and ν be two measures on $\mathcal{A}$. If
(i) $\mu(E) = \nu(E),\ \forall\ E \in \mathcal{E}$.
(ii) $\mu(\Omega) = \nu(\Omega) < \infty$ or $\exists\ (E_n)_{n \geqslant 1} \in \mathcal{E}$ such that $\mu(E_n) = \nu(E_n) \infty$ and $E_n \uparrow \Omega$.
then $\mu = \nu$.

1.7 Product Measure and Fubini's Theorem

So far, we have only considered the one-dimensional Lebesgue measure. In fact, one way to construct the two-dimensional Lebesgue measure is to view it as the product of two one-dimensional Lebesgue measures:

$$\lambda^2((a_1, b_1] \times (a_2, b_2]) = (b_1 - a_1)(b_2 - a_2).$$

More generally, intuitively, we have

$$\lambda^2(A \times B) = \lambda(A) \cdot \lambda(B)$$

where $A, B \in \mathcal{B}(\mathbb{R})$. It is worth noting that

$$\mathcal{R}\left(\mathbb{R}^2\right) := \left\{E \subset \mathbb{R}^2 : E = A \times B,\ A, B \in \mathcal{B}(\mathbb{R})\right\}$$

is not a σ-algebra, because

$$((a_1, b_1] \times (a_2, b_2])^c \notin \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}).$$

Thus, we first need to construct the Borel σ-algebra on $\mathbb{R}^2$, and the most natural idea is to consider the σ-algebra generated by$\mathcal{R}\left(\mathbb{R}^2\right)$.

Definition 1.7.1. Consider two measurable spaces $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$.

a). Let
$$\mathcal{F}_1 \times \mathcal{F}_2 := \{A \times B : A \in \mathcal{F}_1,\ B \in \mathcal{F}_2\}.$$
We intuitively refer to the elements in $\mathcal{A}_1 \times \mathcal{A}_2$ as measurable rectangles.

b) Define
$$\mathcal{F}_1 \otimes \mathcal{F}_2 := \sigma(\mathcal{F}_1 \times \mathcal{F}_2),$$
which is called the product σ-algebra.

The following lemma proves that for measurable functions with respect to the product σ-algebra, their truncations are also measurable.

Lemma 1.7.2. Let f *be a non-negative measurable function on* $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$. Then: 1). Fixing $x \in \Omega_2$, the function $y \mapsto f(x,y)$ is a measurable function on $(\Omega_2, \mathcal{F}_2)$. 2) Let μ be a σ-finite measure on $(\Omega_2, \mathcal{F}_2)$, then

$$x \mapsto \int f(x,y)\mu(\mathrm{d}y)$$

is a measurable function on $(\Omega_1, \mathcal{F}_1)$.

Proof. First step: Prove that when $f = \mathbb{1}E$ and $E \in \mathcal{F}1 \otimes \mathcal{F}2$, the proposition holds. First, there exists $B_j \in \mathcal{F}2$ such that $B_j \uparrow \Omega_2$ and $\mu_2(B_j) < \infty$. Let $E_j := \Omega_1 \times B_j$. Define $\mathcal{D}j \subset \mathcal{F}1 \otimes \mathcal{F}_2$ as follows:

$D \in \mathcal{D}j \Leftrightarrow$ (*a*) $y \mapsto \mathbb{1}D \cap E_j(x,y)$ is $\mathcal{F}2$-measurable, and (*b*) $x \mapsto \int \mathbb{1}D \cap E_j(x,y)\mu(\mathrm{d}y)$ is $\mathcal{F}_1$-measurable.

We then prove $\mathcal{D}_j = \mathcal{F}_1 \otimes \mathcal{F}_2$.

(i). $\mathcal{F}_1 \times \mathcal{F}_2 \subset \mathcal{D}_j$. For $A \in \mathcal{F}_1, B \in \mathcal{F}_2$, obviously we have

$$\mathbb{1}_{(A\times B)\cap E_j}(x,y) = \mathbb{1}_A(x) \cdot \mathbb{1}_{B\cap B_j}(y)$$

$$\int \mathbb{1}_A(x) \cdot \mathbb{1}_{B\cap B_j}(y)\mu(\mathrm{d}y) = \mathbb{1}_A(x) \cdot \mu(B \cap B_j)$$

Therefore, $\mathcal{F}_1 \times \mathcal{F}_2 \subset \mathcal{D}_j$.

(ii). $\mathcal{D}_j$ is a Dynkin class, $\forall j \in \mathbb{N}$.
(1)Obviously $\Omega_1 \times \Omega_2 \in \mathcal{D}_j$;

(2) Let $D \in \mathcal{D}_j$. We have

$$\mathbb{1}_{D^c\cap E_j} = \mathbb{1}_{E_j} - \mathbb{1}_{D\cap E_j}$$

and

$$\int \mathbb{1}_{D^c\cap E_j}(x,y)\mu(dy) = \int \left(\mathbb{1}_{E_j} - \mathbb{1}_{D\cap E_j}\right)(x,y)\mu(dy) = \mu(B_j) - \int \mathbb{1}_{D\cap E_j}(x,y)\mu(dy).$$

Note that $\mu(B_j) < \infty$ and the operations above preserve measurability, so we obtain $D^c \in \mathcal{D}_j$.

(3) Let $(D_k)_{k\in\mathbb{N}} \subset \mathcal{D}_j$ be pairwise disjoint. Denote$D := \bigcup_k D_k$. We have:

$$\mathbb{1}_{D\cap E_j} = \sum_{k\in\mathbb{N}} \mathbb{1}_{D_k\cap E_j}$$

and

$$\int \mathbb{1}_{D\cap E_j}(x,y)\mu_1(dy) = \int \left(\sum_{k\in\mathbb{N}} \mathbb{1}_{D_k\cap E_j}\right)(x,y)\mu(dy) = \sum_{k\in\mathbb{N}} \int \mathbb{1}_{D_k\cap E_j}(x,y)\mu(dy).$$

Since the countable sum of non-negative measurable functions is still measurable, we obtain $D \in \mathcal{D}_j$.

Since $\mathcal{F}1 \times \mathcal{F}2$ is closed under intersection,

$$\mathcal{F}_1 \otimes \mathcal{F}_2 = \sigma(\mathcal{F}_1 \times \mathcal{F}_2) \subset \mathcal{D}_j \subset \mathcal{F}_1 \otimes \mathcal{F}_2,$$

That is, $\mathcal{D}j = \mathcal{F}1 \otimes \mathcal{F}2$. Therefore, (a) and (b) hold for all $E \in \mathcal{F}1 \otimes \mathcal{F}_2$. As $j \to \infty$, we have

$$\mathbb{1}_{E \cap E_j}(x, y) \uparrow \mathbb{1}_E(x, y).$$

According to the monotone convergence theorem, we also have

$$\int \mathbb{1}_{E \cap E_j}(x, y)\mu(dy) \uparrow \int \mathbb{1}_E(x, y)\mu(dy).$$

Since the limit of measurable functions is still measurable, the proposition holds for$f = \mathbb{1}_E$ with $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$.

Second step: Based on the first step and the linearity of integration, the proposition obviously holds for non-negative simple functions. Furthermore, by the monotone convergence theorem, the proposition also holds for non-negative measurable functions. □

Now, we can apply the above lemma to define the product measure.

Theorem 1.7.3. *Let* $(\Omega_1, \mathcal{F}1, \mu 1)$ *and* $(\Omega_2, \mathcal{F}2, \mu 2)$ *be* σ-finite measures*. Then, on* $(\Omega_1 \times \Omega_2, \mathcal{F}1 \otimes \mathcal{F}2)$*, there exists a unique measure* $\mu_1 \otimes \mu_2$ such that

$$\mu_1 \otimes \mu_2(A \times B) = \mu_1(A)\mu_2(B), \quad A \in \mathcal{F}_1, B \in \mathcal{F}_2.$$

Moreover, $\mu_1 \otimes \mu_2$ *is a* σ*-finite measure, and for any* $E \in \mathcal{F}1 \otimes \mathcal{F}2$*, we have*

$$\begin{aligned}\mu_1 \otimes \mu_2(E) &= \int \left(\int \mathbb{1}_E(x, y)\mu_2(\mathrm{d}y) \right) \mu_1(\mathrm{d}x) \\ &= \int \left(\int \mathbb{1}_E(x, y)\mu_1(\mathrm{d}x) \right) \mu_2(\mathrm{d}y).\end{aligned}$$

Proof. For $E \in \mathcal{F}1 \otimes \mathcal{F}2$*, define*

$$\mu_1 \otimes \mu_2(E) := \int \left(\int \mathbb{1}_E(x, y)\mu_2(\mathrm{d}y) \right) \mu_1(\mathrm{d}x).$$

According to Lemma 1.7.2, $\mu_1 \otimes \mu_2$ is well-defined. Clearly, $\mu_1 \otimes \mu_2(\varnothing) = 0$. Let $E_1, E_2, \ldots \in \mathcal{F}1 \otimes \mathcal{F}2$ be pairwise disjoint. By Corollary 1.5.4, we have

$$\begin{aligned}\mu_1 \otimes \mu_2(\cup_{j=1} E_j) &= \int \left(\int \mathbb{1}_{\cup_{j=1} E_j}(x, y)\mu_2(\mathrm{d}y) \right) \mu_1(\mathrm{d}x) \\ &= \int \left(\sum_{j=1} \int \mathbb{1}_{E_j}(x, y)\mu_2(\mathrm{d}y) \right) \mu_1(\mathrm{d}x) \\ &= \sum_{j=1} \int \left(\int \mathbb{1}_{E_j}(x, y)\mu_2(\mathrm{d}y) \right) \mu_1(\mathrm{d}x) = \sum_{j=1} \mu_1 \otimes \mu_2(E_j).\end{aligned}$$

Thus, $\mu_1 \otimes \mu_2$ is a measure on $\mathcal{F}1 \otimes \mathcal{F}2$. Furthermore, there exist sets $A_j \in \mathcal{F}1$ such that $Aj \uparrow \Omega_1$ and $\mu_1(A_j) < \infty$, and sets $B_j \in \mathcal{F}2$ such that $Bj \uparrow \Omega_2$ and $\mu_2(B_j) < \infty$. Therefore, $A_j \times B_j \uparrow \Omega_1 \times \Omega_2$ and $\mu_1 \otimes \mu_2(A_j \times B_j) < \infty$, implying that $\mu_1 \otimes \mu_2$ is σ-finite. Since $\mathcal{F}1 \times \mathcal{F}2$ is $\cap$-stabled, the uniqueness of $\mu_1 \otimes \mu_2$ follows from Theorem 1.6.3. □

We now provide the Fubini's theorem.

Theorem 1.7.4. *Let* $(\Omega_1, \mathcal{F}1, \mu 1)$ *and* $(\Omega_2, \mathcal{F}2, \mu 2)$ *be* σ-finite measures*, and let* f *be a non-negative measurable function on* $(\Omega_1 \times \Omega_2, \mathcal{F}1 \otimes \mathcal{F}2)$*. If* f *is non-negative or integrable, then the double integral equals the iterated integrals, i.e.,*

$$\begin{aligned}\int_{\Omega_1\times\Omega_2} f\mathrm{d}\mu_1\otimes\mu_2 &= \int_{\Omega_1}\left(\int_{\Omega_2} f(x,y)\mu_2(\mathrm{d}y)\right)\mu_1(\mathrm{d}x)\\ &= \int_{\Omega_2}\left(\int_{\Omega_1} f(x,y)\mu_1(\mathrm{d}x)\right)\mu_2(\mathrm{d}y).\end{aligned}$$

Proof. According to Theorem 1.7.3, when $f = \mathbb{1}E$ and $E \in \mathcal{F}1 \otimes \mathcal{F}_2$, the proposition holds. By the linearity of the integral, the proposition clearly holds for non-negative simple functions. Moreover, by the monotone convergence theorem, the proposition also holds for non-negative measurable functions.

Next, consider an integrable f. For its positive part, since $f = f^+ - f^-$ and f^+, we have

$$\int_{\Omega_1}\left(\int_{\Omega_2} f^+(x,y)\mu_2(\mathrm{d}y)\right)\mu_1(\mathrm{d}x) = \int_{\Omega_1\times\Omega_2} f^+\mathrm{d}\mu_1\otimes\mu_2 < \infty.$$

It is not difficult to see that $x \mapsto \int_{\Omega_2} f^+(x,y)\mu_2(\mathrm{d}y)$ is finite almost everywhere with respect to μ_1 and is integrable. Thus,

$$\begin{aligned}x \mapsto &\int_{\Omega_2} f(x,y)\mu_2(\mathrm{d}y)\\ &= \int_{\Omega_2}\left(\int_{\Omega_1} f^+(x,y)\mu_1(\mathrm{d}x)\right)\mu_2(\mathrm{d}y) - \int_{\Omega_2}\left(\int_{\Omega_1} f^-(x,y)\mu_1(\mathrm{d}x)\right)\mu_2(\mathrm{d}y)\end{aligned}$$

is thus well-defined (taking real values) and integrable almost everywhere with respect to μ_1. Therefore,

$$\begin{aligned}\int_{\Omega_1\times\Omega_2} f\mathrm{d}\mu_1\otimes\mu_2 &= \int_{\Omega_1\times\Omega_2} f^+\mathrm{d}\mu_1\otimes\mu_2 - \int_{\Omega_1\times\Omega_2} f^-\mathrm{d}\mu_1\otimes\mu_2\\ &= \int_{\Omega_1}\left(\int_{\Omega_2} f^+(x,y)\mu_2(\mathrm{d}y)\right)\mu_1(\mathrm{d}x) - \int_{\Omega_1}\left(\int_{\Omega_2} f^-(x,y)\mu_2(\mathrm{d}y)\right)\mu_1(\mathrm{d}x)\\ &= \int_{\Omega_2}\left(\int_{\Omega_1} f(x,y)\mu_1(\mathrm{d}x)\right)\mu_2(\mathrm{d}y).\end{aligned}$$

The other equality can be proven in a completely similar manner. The proposition is proven. □

Example 1.7.5. Let u be a non-negative measurable function on the measure space $(\Omega, \mathcal{A}, \mu)$. Then,

$$\int_\Omega u d\mu = \int_{(0,\infty)} \mu(u \geqslant t)\lambda(dt) \ \in \ [0,\infty].$$

Proof. We have

$$\begin{aligned}\int_\Omega u(x)\mu(dx) &= \int_\Omega (\int_0^{u(x)} 1dt)\mu(dx)\\ &= \int_\Omega \int \mathbb{1}_{[0,u(x)]}dt\mu(dx)\\ &= \int_\Omega \int_{[0,\infty)} \mathbb{1}_{[0,u(x)]}(t)\lambda(dt)\mu(dx)\\ &= \int_{[0,\infty)} \int_\Omega \mathbb{1}_{[0,u(x)]}(t)\mu(dx)\lambda(dt)\\ &= \int_{\Omega\times[0,\infty)} \mathbb{1}_E(x,t)(\mu\lambda)(dx,dt),\end{aligned}$$

of which

$$\begin{aligned}\mathbb{1}_E(x,t) &= \mathbb{1}_{\{(y,s)\in\Omega\times[0,\infty):u(y)\geqslant s\}}(x,t)\\ &= \begin{cases} 1, & u(x)\geqslant t\\ 0, & u(x)<t\end{cases}\\ &= \mathbb{1}_{[0,u(x)]}(t) = \mathbb{1}_{[t,\infty)}(u(x))\\ &\Rightarrow \mu(u\geqslant t) = \int \mathbb{1}_E(x,t)\mu(dx)\end{aligned}$$

It still needs to be verified: is $E \in \mathcal{A}\otimes\mathcal{B}(\mathbb{R}+)$? Notice that: $U(x,t) := (u(x),t)$ is $\mathcal{A}\otimes\mathcal{B}(\mathbb{R}+)$-measurable:

$$U^{-1}(A\times I) = u^{-1}(A)\times I \in \mathcal{A}\times\mathcal{B}(\mathbb{R}_+) \subset \mathcal{A}\otimes\mathcal{B}(\mathbb{R}_+).$$

Thus:

$$E = \{(y,\ s)\ :\ u(y)\geqslant s\} = \{(y,s)\ :\ u(y)-s\geqslant 0\} \in \mathcal{A}\otimes\mathcal{B}(\mathbb{R}_+).$$

□

Exercises

1. Let $\Omega_1 = \Omega_2 = \mathbb{N}$, $\mathcal{F}_1 = \mathcal{F}_2 = 2^{\mathbb{N}}$, and $\mu_1 = \mu_2$ be the counting measure, i.e.,

   $$\mu_i(A) = \#(A).$$

   Define $f(n,n) = n$, $f(n,n+1) = -n$, $n = 1,2,\ldots$, and $f(i,j) = 0$ if $j \neq i$ or $j \neq i+1$. Prove that

   $$\int_{\Omega_1}\int_{\Omega_2} f d\mu_2 d\mu_1 = 0, \quad \int_{\Omega_2}\int_{\Omega_1} f d\mu_1 d\mu_2 = \infty.$$

   Does this example contradict the Fubini Theorem? Please explain.

1.8 Comparison of Lebesgue Integral and Riemann Integral

Theorem 1.8.1 (Existence and Uniqueness of the One-Dimensional Lebesgue Measure)**.** *There exists a unique measure λ on $(\mathbb{R},\mathcal{B}(\mathbb{R}))$ such that*

$$\lambda((a,b]) = b-a \quad \forall a,b\in\mathbb{R},\ a\leqslant b.$$

Proof. Let λ be a measure that satisfies the conditions of the theorem. Then

$$\lambda((-n,n]) = n-(-n) = 2n < \infty \quad \forall n,$$

so λ is σ-finite on the collection $\varepsilon :=$(a, b] : a, b $\in \mathbb{R}, a \leqslant b\}$, and ε is closed under intersections and is the generator of $\mathcal{B}(\mathbb{R})$. By the uniqueness of the measure, we conclude the uniqueness of λ.

For existence, we provide a probabilistic proof in later chapters. The main idea is to prove the existence of an i.i.d. sequence of random variables $(X_n)_{n\geqslant 1}$, where X_n is uniformly distributed on $[0,1]$. □

Consider $I = [a,b] \subset \mathbb{R}$. Let $t = (t^n)_{n\in\mathbb{N}}$be a sequence of progressively finer partitions of$I$, where $t^n = (t_i^n)_{i=0,\dots,n}$, $a = t_0^n < t_1^n < \cdots < t_n^n = b$, $\max_{i=l,\dots,n}\{t_i^n - t_i^n\} \xrightarrow{n\to\infty} 0$. Here, we assume that $\{t_0^n,\dots,t_n^n\} \subset \{t_0^{n+1}, t_{n+1}^{n+1}\}$. For $f : I \to \mathbb{R}$ and $n \in \mathbb{N}$, we define

$$U_n^t(f) := \sum_{i=1}^n \left(t_i^n - t_{i-1}^n\right) \inf\{f(t) :\ t \in \left[t_{i-1}^n, t_i^n\right)\}.$$
$$O_n^t(f) := \sum_{i=1}^n \left(t_i^n - t_{i-1}^n\right) \sup\{f(t) :\ t \in \left[t_{i-1}^n, t_i^n\right)\}.$$

We say that f is Riemann integrable if there exists a partition t such that $\lim_{n\to\infty} U_n^t(f) = \lim_{n\to\infty} O_n^t(f)$ and $\lim_{n\to\infty} U_n^t(f)$ is finite. It can be proven that, in this case, the limit $\lim_{n\to\infty} U_n^t(f)$ does not depend on the choice of partition t. We then define the Riemann integral of f on I as

$$\int_a^b f(x)dx := \lim_{n\to\infty} U_n^t(f) = \lim_{n\to\infty} O_n^t(f)$$

Theorem 1.8.2. *Assume that $f : I \longrightarrow \mathbb{R}$ is Riemann integrable on $I = [a,b]$. Then f is Lebesgue integrable on I with respect to the Lebesgue measure λ, and*

$$\int_I f d\lambda = \int_a^b f(x)dx.$$

Proof. Exercise. □

Exercises

1. Please prove Theorem 1.8.2.

Chapter 2

Random Variables

In this chapter, we always assume that $(\Omega, \mathcal{A}, \mathbb{P})$ is a probability space. If the set $A \in \mathcal{A}$, then we call A an event. $\mathbb{P}[A]$ represents the probability that event A occurs.

Definition 2.0.1 (Random Variable)**.** If $X : \Omega \to \mathbb{R}$ is $\mathcal{A}$-measurable, we call X a (real-valued) random variable.

From now on, we will abbreviate events like $(\omega \in \Omega : X(\omega) > a)$ as $(X > a)$.

Definition 2.0.2 (Distribution and Distribution Function)**.** Let X be a random variable.
(i) We call the probability measure $\mathbb{P}_X := \mathbb{P} \circ X^{-1}$ the distribution of X. If $\mu = \mathbb{P}_X$, we write $X \sim \mu$ and say that the distribution of X is μ. If Y is another random variable and $\mathbb{P}_X = \mathbb{P}_Y$, we say that X and Y are identically distributed and write $X \overset{d}{=} Y$.
(ii) We define $F_X : \mathbb{R} \to [0, 1]$ as

$$F_X(x) := \mathbb{P}_X((-\infty,\ x]) = \mathbb{P}(X \leqslant x)\ .$$

as the distribution function of X.

Proposition 2.0.3. *For any random variable X, the distribution function F_X satisfies the following properties:*

(a) F_X is non-decreasing.

(b) F_X is right-continuous and has left limits.

(c) $\lim_{x\to\infty} F_X(x) = 1$ and $\lim_{x\to-\infty} F_X(x) = 0$.

Proof. We only prove the first part of (b). If $x_n \downarrow x$, then $(X \leqslant x_n) \downarrow (X \leqslant x)$. Hence, $\mathbb{P}(X \leqslant x_n) \downarrow \mathbb{P}(X \leqslant x)$, because $\mathbb{P}$ is a measure. □

Proposition 2.0.4. *For every function $F : \mathbb{R} \to [0, 1]$ that satisfies the properties (a)-(c) of Proposition 2.0.3, there exists a random variable X such that the distribution function of X is equal to F.*

Proof. Choose $\Omega := (0, 1)$, $\mathcal{A} := \mathcal{B}(\mathbb{R})|_\Omega$, and the Lebesgue measure $\mathbb{P}$ on $(\Omega, \mathcal{A})$. Define a generalized inverse of F by $F^{-1}(t) := \inf\{x \in \mathbb{R} : F(x) \geqslant t\},\ t \in (0, 1)$. It can be observed

that $F^{-1}(t) \leqslant x \iff t \leqslant F(x)$. In particular, we have $\{t : F^{-1}(t) \leqslant x\} = (0, F(x)] \bigcap (0,1)$, so $F^{-1} : (\Omega, \mathcal{A}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable, and $\mathbb{P}(\{t : F^{-1}(t) \leqslant x\}) = F(x)$. Hence, $X = F^{-1}$ is the desired random variable.

□

Example 2.0.5. (i) Let $p \in [0,1]$, and if $\mathbb{P}(X=1) = p$, $\mathbb{P}(X=0) = 1-p$, then the distribution of X is denoted by Ber_p, and it is called the Bernoulli distribution with parameter p.
(ii) Let $p \in [0,1]$, $n \in \mathbb{N}$. Consider the random variable X: $\Omega \to \{0, \cdots, n\}$, if

$$\mathbb{P}(X=k) = \left(\begin{array}{c} n \\ k \end{array} \right) p^k (1-p)^{n-k},\ k = 0, \cdots, n,$$

then the distribution of X is called the binomial distribution with parameters n and p.
(iii) Let $p \in (0,1]$, $X:\ \Omega \to \mathbb{N}_0$. If $\mathbb{P}(X=n) = p \cdot (1-p)^{n-1}, \forall n \in \mathbb{N}_0$, then X is said to have the geometric distribution with parameter p, denoted by $\gamma_p := \mathbb{P}_x$.
(iv) Let $\lambda \in [0,\ \infty)\ ,\ X : \Omega \to \mathbb{N}_0\,,$ If

$$\mathbb{P}(X=n) = e^{-\lambda} \frac{\lambda^n}{n!}\ \forall n \in \mathbb{N}_0,$$

then the distribution of X is called the Poisson distribution with parameter λ and is denoted by $\text{Poi}_\lambda := \mathbb{P}_x$.
(v) On the uniform interval $[a,\ b]$, if X satisfies

$$\mathbb{P}(X \in A) = \int_A \frac{1}{b-a} 1_{[a,b]}(x)\ dx,\ \forall A \in \mathcal{B}(\mathbb{R}),$$

then the distribution of X, $U_{[a,b]} := \mathbb{P}_x$, is called the uniform distribution on $[a,b]$.
(vi) Let $\mu \in \mathbb{R},\ \sigma^2 > 0,\ X : \Omega \to \mathbb{R}$, if

$$\mathbb{P}(X \in A) = \frac{1}{2\pi\sigma^2} \int_A \exp(-\frac{(t-\mu)^2}{2\sigma^2}) dt,\ \forall A \in \mathcal{B}(\mathbb{R}),$$

then we say that the distribution of X, $N_{(\mu,\sigma^2)} := \mathbb{P}(X)$, is the normal distribution with parameters μ and σ^2.
(vii) Assume $X \geqslant 0$, $\theta > 0\,,$ if

$$\mathbb{P}(X \leqslant x) = \mathbb{P}(X \in [0,\ x]) = \int_0^x \theta e^{-\theta t} dt \text{ fur } x \geqslant 0,$$

then the distribution of X, Exp_θ, is called the exponential distribution with parameter θ.
(viii) Let $X : \Omega \to \mathbb{R},\ a > 0$, if

$$\mathbb{P}(X \in A) = \int_A \frac{1}{a\pi} \cdot \frac{1}{1 + (\frac{x}{a})^2} dx,\ \forall A \in \mathcal{B}(\mathbb{R}),$$

then we say that X has the Cauchy distribution with parameter a, denoted by $\text{Cau}_a := \mathbb{P}_x$.

Definition 2.0.6 (Expectation)**.** Let X be a random variable. The integral of X over Ω is called the expectation of X, and it is denoted by $E[X]$. $E[X] = \int_\Omega X d\mathbb{P}$. The integral of X over A is denoted by $E[X; A] := \int_A X d\mathbb{P}$.

Proposition 2.0.7. *Assume X is a random variable, and $g : \mathbb{R} \to \mathbb{R}$ is a $\mathcal{B}(\mathbb{R})$-measurable function. If g is bounded and non-negative, then $E[g(X)] = \int_{\mathbb{R}} g(x)\mathbb{P}_x(dx)$.*

Proof. If $g = 1_A$, $A \in \mathcal{B}(\mathbb{R})$, then clearly $E[1_A(X)] = \mathbb{P}(x \in A) = \mathbb{P}_x(A)$.

By the linearity of the integral, the above equality can be extended to simple functions. Then, by the monotone convergence theorem, we can derive the above equality for any non-negative measurable function or any non-negative bounded function g.

□

Definition 2.0.8 (Variance)**.** Consider a random variable X, $E|X| < \infty$. The variance of X is defined as $VarX = E[(X - EX)^2]$.
It can be observed that $VarX = E[X^2 - 2XE[X] + (E[X])^2] = E[X^2] - (E[X])^2$.

Proposition 2.0.9. *For a random variable $X \geqslant 0$ and any $p > 0$, $E[X^p] = \int_0^\infty p \cdot t^{p-1}\mathbb{P}(X > t)dt$. If we replace $\mathbb{P}(X > t)$ with $\mathbb{P}(X \geqslant t)$, we can see that the above equality still holds.*

Proof. By Fubini's theorem, we have

$$\begin{aligned}
\int_0^\infty p \cdot t^{p-1}\mathbb{P}(X > t)dt &= \int_0^\infty p \cdot t^{p-1}\mathbb{E}\left[1_{(t,\infty)}(X(\omega))\right] dt \\
&= \mathbb{E}\left[\int_0^\infty p \cdot t^{p-1}1_{(t,\infty)}(X(\omega))dt\right] \\
&\overset{*}{=} \mathbb{E}\left[\int_0^{X(\omega)} pt^{p-1}dt\right] \\
&= \mathbb{E}\left[X^p\right]
\end{aligned}$$

For the above step (*), let $E := \{(\omega, t) : X(\omega) > t\}$. Then

$$\begin{aligned}
\mathbf{1}_E(\omega, t) &= \begin{cases} 1, & X(\omega) > t \\ 0, & X(\omega) \leqslant t \end{cases} \\
&= \mathbf{1}_{[0,x)}(t) \\
&= \mathbf{1}_{(t,\infty)}(x).
\end{aligned}$$

Therefore, it suffices to verify that $E \in \mathcal{A} \otimes \mathcal{B}(\mathbb{R}_+)$. Define $\tilde{X}(\omega, t) := X(\omega)$ and $\tilde{Y}(\omega, t) := t$. It is easy to see that both are $\mathcal{A} \otimes \mathcal{B}(\mathbb{R}_+)$-measurable functions. Therefore,

$$E = \{\tilde{X} > \tilde{Y}\} \in \mathcal{A} \otimes \mathcal{B}(\mathbb{R}_+).$$

□

Proposition 2.0.10 (Markov and Chebyshev Inequalities)**.** *For constants $a, p > 0$*
(i) Consider a non-negative random variable $X \geqslant 0$. Then we have the Markov inequality: $\mathbb{P}(X \geqslant a) \leqslant \frac{\mathbb{E}[X]}{a}$.
(ii) Consider a non-negative random variable $X \geqslant 0$. Then we have the Chebyshev inequality: $\mathbb{P}(|X| \geqslant a) \leqslant \frac{\mathbb{E}[|X|^p]}{a^p}$.

Proof. (i) Since X is non-negative, we have $a \cdot 1_{\{x \geqslant a\}} \leqslant X$, hence $a \cdot \mathbb{P}(X \geqslant a) = \mathbb{E}[a \cdot 1_{\{X \geqslant a\}}] \leqslant E[X]$, which gives the Markov inequality.
(ii) By the Markov inequality, we have $\mathbb{P}(|X| \geqslant a) = \mathbb{P}(|X|^p \geqslant a^p) \leqslant \frac{E[|X|^p]}{a^p}$. □

Definition 2.0.11. [Convex Function] Let $g : I \to \mathbb{R}$, where $I \subset \mathbb{R}$ is an open interval (bounded or unbounded). We say that g is convex if

$$g(ax + (1-a)y) \leqslant ag(x) + (1-a)g(y), \quad \forall x, y \in I, \ \forall a \in [0,1].$$

Lemma 2.0.12. *Let $I \subset \mathbb{R}$ be an open interval, and let $g : I \to \mathbb{R}$ be a convex function. Then for any $a \in I$, we can find a real number b such that*

$$g(x) \geqslant g(a) + b(x-a), \quad \forall x \in I.$$

Proof. Let $0 < h_1 < h_2$. By the convexity of g, if $x \in I$ and $(x - h_2, x + h_2) \subset I$, then

$$g(x \pm h_1) \leqslant \frac{(h_2 - h_1)}{h_2} g(x) + \frac{h_1}{h_2} g(x \pm h_2),$$

thus,

$$\frac{1}{h_1}[g(x + h_1) - g(x)] \leqslant \frac{1}{h_2}[g(x + h_2) - g(x)] \tag{2.0.1}$$

and

$$-\frac{1}{h_2}[g(x - h_2) - g(x)] \leqslant -\frac{1}{h_1}[g(x - h_1) - g(x)]. \tag{2.0.2}$$

Similarly, if $h, h' > 0$, and $x \in I$ such that $(x - h, x + h') \subset I$, then

$$g(x) \leqslant \frac{h'}{h + h'} g(x - h) + \frac{h}{h + h'} g(x + h'),$$

so that

$$\frac{g(x-h) - g(x)}{-h} \leqslant \frac{g(x + h') - g(x)}{h'}. \tag{2.0.3}$$

According to (2.0.1) and (2.0.2), the right derivative $g_+{}'$ and the left derivative $g_-{}'$ both exist on I, and according to (2.0.3), both must be finite. Furthermore, from (2.0.1), (2.0.2), and (2.0.3), we have

$$g'_+(x) = \inf_{y > x} \frac{g(y) - g(x)}{y - x}, \quad g'_-(x) = \sup_{y < x} \frac{g(x) - g(y)}{x - y}$$

and

$$g'_-(x) \leqslant g'_+(x)$$

if $x_1 < x_2$, then

$$g'_+(x_1) \leqslant \frac{g(x_2) - g(x_1)}{x_2 - x_1} = \frac{g(x_1) - g(x_2)}{x_1 - x_2} \leqslant g'_-(x_2). \tag{2.0.4}$$

Therefore, $g_+{}'$ and $g_-{}'$ are monotonically non-decreasing on I.

According to(2.0.4), if $y \geqslant x$, then

$$g(y) \geqslant g(x) + (y - x) g'_+(x);$$

if $y < x$, then

$$g(y) \geqslant g(x) + (y - x)g'_-(x).$$

Now, select $a_x \in \left[g'_-(x), g'_+(x)\right]$, then

$$g(y) \geqslant g(x) + (y - x)a_x \quad \forall y \in I.$$

The lemma is proved. □

Proposition 2.0.13 (Jensen's Inequality)**.** *Let $X : \Omega \to I$ be a random variable taking values in the open interval I, where $\mathbb{E}|X| < \infty$, and let $g : I \to \mathbb{R}$ be a convex function. Then $\mathbb{E}[g(x)]$ necessarily exists and*

$$g(E[X]) \leqslant \mathbb{E}[g(X)].$$

Proof. For any $a \in I$, there exists $b \in \mathbb{R}$ such that

$$g(x) \geqslant g(a) + b(x - a) \quad \forall x \in I.$$

Therefore,

$$g(x) \geqslant g(a) + b(x - a)$$

On the other hand,

$$\begin{aligned} f^-(x) &\leqslant (g(a) + b(x - a))^- \\ &\leqslant |g(a)| + |b||x| + |ab| \end{aligned}$$

Thus, $\mathbb{E}[g^-(X)] < \infty$ because $\mathbb{E}[|X|] < \infty$ and $\mathbb{P}(\Omega) = 1$. Therefore, $\mathbb{E}[g(X)]$ exists. If $\mathbb{E}\left[(g(x))^+\right] = \infty$, the conclusion of the proposition obviously holds. Now assume $\mathbb{E}\left[(g(x))^+\right] < \infty$. Then, by the monotonicity and linearity of the integral and since $\mathbb{P}(\Omega) = 1$, we have

$$\mathbb{E}[g(x)] \geqslant g(a) + b(E[x] - a),$$

Choosing $a = \mathbb{E}[x]$ completes the proof of the proposition. □

Proposition 2.0.14 (Random Vector)**.** *Let $X_1, X_2, \ldots, X_n$ be random variables. Define$X := (x_1, x_2, x_n)$. Then $X : \Omega \longrightarrow \mathbb{R}^n$ is $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable. Here,$\mathcal{B}(\mathbb{R}^n) = \mathcal{B}(\mathbb{R}) \otimes \cdots \otimes \mathcal{B}(\mathbb{R})$. In this case, we call X a random vector and the image measure (defined on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$) given by$\mathbb{P}_x := \mathbb{P} \circ X^{-1}$ the joint distribution of $X_1, \ldots, X_n$.*

Proof. Let $X_1, X_2, \ldots, X_n$ be random variables. For all $\forall A_1, A_2, \ldots, A_n \in \mathcal{B}(\mathbb{R})$, we have

$$\begin{aligned} X^{-1}(A_1 \times A_2 \times \ldots \times A_n) &= \left\{X_1^{-1}(A_1), X_2^{-1}(A_2), \ldots, X_n^{-1}(A_n)\right\} \\ &= \left\{\omega \in \Omega : \omega \in X^{-1}(A_1), \cdots, X_n^{-1}(A_n)\right\} \\ &= X_1^{-1}(A_1) \cap X_2^{-1}(A_2) \cap \cdots \cap X_n^{-1}(A_n) \in \mathcal{A} \end{aligned}$$

since $X_1, X_2, \ldots, X_n$ are random variables. Let $\varepsilon := \{A_1 \times A_2 \times \cdots \times A_n : A_i \in \mathcal{B}(\mathbb{R})\ i = 1, \ldots, n\}$. Then ε is the generating set of $B(\mathbb{R}^n)$and is closed under intersections. However, $X^{-1}(\varepsilon) \subset \mathcal{A}$. By Lemma 1.2.1, $X : \Omega \longrightarrow \mathbb{R}^n$ is $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable. □

Remark 2.0.15. (i) Let $P_i : \mathbb{R}^n \to \mathbb{R}$ be the projection of a vector in $\mathbb{R}^n$ onto its i-th component,

$$x = (x_1, x_2, \ldots, x_n) \xrightarrow{P_i} x_i$$

Consider a mapping $h : \Omega \to \mathbb{R}^n$ and define $h_i := P_i \circ h$. Then h is $\mathcal{A}/\mathcal{B}(\mathbb{R}^n)$-measurable if and only if each h_i is $\mathcal{A}/\mathcal{B}(\mathbb{R})$-measurable for all $i = 1, \ldots, n$.

Proof: Since the composition of measurable functions is measurable, we only need to observe that $P_i : \mathbb{R}^n \longrightarrow \mathbb{R}$ is$\mathcal{B}(\mathbb{R}^n)/\mathcal{B}(\mathbb{R})$-measurable.
(ii) Let $X = (X_1, \ldots, X_n)$ be a random vector and $g : \mathbb{R}^n \to \mathbb{R}$ be measurable with respect to $\mathcal{B}(\mathbb{R}^n)/\mathcal{B}(\mathbb{R})$. If g is bounded or non-negative, then

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}^n} g(x)\mathbb{P}_X(dx).$$

To prove this, we only need to apply the standard method: first verify it for indicator functions of Borel sets, then pass to simple functions, and finally use the monotone convergence theorem or the dominated convergence theorem to handle the general case.

Exercises

1. Let F be the distribution function of some random variable. Prove that
$$\int_{\mathbb{R}} (F(x+c) - F(x))dx = c, \quad \forall c \in \mathbb{R}.$$

2. Let F be the distribution function of some random variable. Prove that F has at most countably many discontinuities.

3. Let ξ and η be random variables with distribution functions F and G, respectively. Prove that $\xi \overset{d}{=} \eta$ if and only if $F = G$.

4. Let X be a random variable. Suppose $F(x) = P(X \leqslant x)$ is a continuous function on $\mathbb{R}$. Prove that $Y = F(X)$ has the uniform distribution on $(0, 1)$, i.e., for $y \in [0, 1]$, $P(Y \leqslant y) = y$.

5. Let X be a random variable with $X \geqslant 0$. Prove that
$$\lim_{y\to\infty} yE(1/X; X > y) = 0, \quad \lim_{y\downarrow 0} yE(1/X; X > y) = 0.$$

6. Let X and Y be random variables. Prove that if $p, q \in (1, \infty)$ satisfy $1/p + 1/q = 1$, then
$$E|XY| \leqslant \|X\|_p \|Y\|_q,$$
where $\|X\|_r = (E|X|^r)^{1/r}$, $r \in (1, \infty)$.

Chapter 3

Independence

3.1 Definitions and Basic Properties

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space.

Definition 3.1.1 (Generated σ-Algebra)**.** (i) Let $X : \Omega \to \mathbb{R}$ be a random variable. It is easy to see that

$$X^{-1}(\mathcal{B}(\mathbb{R})) := \left\{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\right\}$$

is the smallest σ-algebra that makes X measurable. We denote by $\sigma(X) := X^{-1}(B(\mathbb{R}))$ the σ-algebra generated by X on Ω. (ii) Let I be an index set. For each $i \in I$, let $X_i : \Omega \to \mathbb{R}$ be a random variable, and

$$\sigma\left(X_i, i \in I\right) := \sigma\left(\bigcup_{i \in I} \sigma\left(X_i\right)\right) = \sigma\left(\bigcup_{i \in I} X_i^{-1}(B(\mathbb{R}))\right).$$

is the σ-algebra generated by $(X_i, i \in I)$ on Ω. It is the smallest σ-algebra that makes all X_i measurable.

Definition 3.1.2 (Independence)**.** (i) A family of events$(A_i)_{i \in I}$, $A_i \in \mathcal{A}$, is mutually independent if and only if

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}\left(A_j\right) \tag{3.1.1}$$

holds for any $J \subset\subset I$. Here, $J \subset\subset I$ means $J \subset I$ and $\#(J) < \infty$. (vel Koringut)
(ii) A family of collections of sets $(\mathcal{C}_i)_{i \in I}$, $\mathcal{C}_i \subset\subset \mathcal{A}$, is mutually independent if and only if 3.1.1 holds for any $J \subset\subset I$ and $A_j \in \mathcal{C}_j$ for $j \in J$.
(iii) A family of random variables $(X_i)_{i \in I}$ is mutually independent if and only if $(\sigma\left(X_i\right))_{i \in I}$ is mutually independent.

Proposition 3.1.3. *(Independence and Product Measure) A family of random variables* $(X_i)_{i \in I}$ *is mutually independent if and only if for every*$J = \{j_1, j_2, \cdots, j_n\} \subset\subset I$, $\mathbb{P}_{\left(x_{i_1}, x_{i_2}, \ldots, x_{i_n}\right)} = \mathbb{P}_{x_{i_1}} \otimes \mathbb{P}_{x_{i_2}} \otimes \cdots \otimes \mathbb{P}_{x_{i_n}}$, *that is, the joint distribution of any finite subfamily equals the product of their individual distributions.*

Proof. By definition, $(X_i)_{i\in I}$ are mutually independent if and only if $\forall J = \{j_1, j_2, \cdots, j_n\} \subset\subset I, A_{j_k} \in \mathcal{B}(\mathbb{R}), k = 1, 2, \ldots, n, \mathbb{P}(X_{j_1} \in A_{j_1}, X_{j_2} \in A_{j_2}, \cdots, X_{j_n} \in A_{j_n}) = \prod_{k=1}^{n} P(X_{j_k} \in A_{j_k})$. Since $\mathbb{P}(X_{j_k} \in A_{j_k}) = \mathbb{P}_{X_{j_k}}(A_{j_k})$ and $\mathbb{P}(X_{j_1} \in A_{j_1}, X_{j_2} \in A_{j_2}, \cdots, X_{j_n} \in A_{j_n}) = \mathbb{P}_{X_{j_1},\cdots,X_{j_n}}(A_{j_1} \times \cdots \times A_{i_n})$, by Fubini's theorem, we can obtain that

$$\mathbb{P}_{(x_{j_1}\cdots,x_{j_n})}(A_{j_1}, \cdots A_{j_n}) = \mathbb{P}_{x_j}(A_{j_1}) \cdot \mathbb{P}_{x_j}(A_{j_2}) \ldots \mathbb{P}_{x_{j_n}}(A_{j_n})$$

Since $A_{j_1}, \cdots, A_{j_n}$ are arbitrary, $\mathbb{P}_{(x_{i1},x_{i2},\ldots,x_{in})} = \mathbb{P}_{x_{i_1}} \otimes \mathbb{P}_{x_{i2}} \otimes \cdots \otimes \mathbb{P}_{x_{in}}$. □

Lemma 3.1.4 (Functions of Independent Random Variables)**.** *If $(X_i)_{i\in I}$ are mutually independent and $\forall i \in I, \varphi_i : \mathbb{R} \longrightarrow \mathbb{R}$ is measurable, then $(\varphi_i(X_i))_{i\in I}$ are also mutually independent.*

Proof. The random variable $\varphi_i(X_i)$ is measurable with respect to $\sigma(X_i)$, so $\forall i \in I$, $\sigma(\varphi_i(X_i)) \subset \sigma((X_i))$. Since $(\sigma(X_i))_{1\in I}$ are mutually independent, it follows that $(\varphi_i(X_i))_{i\in I}$ are also mutually independent. □

Proposition 3.1.5. *There exist independent random variables X, Y such that $\mathbb{E}|X| < \infty$ and $\mathbb{E}|Y| < \infty$. Then $\mathbb{E}|XY| < \infty$ and $\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$.*

Proof. According to Proposition3.1.3, $\mathbb{P}_{(x,y)} = \mathbb{P}_x \otimes \mathbb{P}_y$. From Fubini's theorem, it follows that

$$\begin{aligned}
\int_{\mathbb{R}^2} |xy| \mathbb{P}_{(x,y)}(dx, dy) &= \int_{\mathbb{R}^2} |xy| \mathbb{P}_x \otimes \mathbb{P}_y(dx, dy) \\
&= \int_{\mathbb{R}} \mathbb{P}_x(dx) \cdot \left(\int_{\mathbb{R}} |xy| \mathbb{P}_y(dy) \right) \\
&= \int_{\mathbb{R}} |x| \mathbb{P}_x(dx) \cdot \left(\int_{\mathbb{R}} |y| \mathbb{P}_y(dy) \right) \\
&= \mathbb{E}|x| \cdot \mathbb{E}|y| \quad < \infty
\end{aligned}$$

Therefore, the function $(x, y) \mapsto xy$ on $\mathbb{R}^2$ is integrable with respect to $\mathbb{P}_{(x,y)}$,

$$\mathbb{E}|xy| = \int_{\mathbb{R}^2} |xy| \mathbb{P}_{(x,y)}(dx_1 dy) < \infty.$$

Similarly, according to Fubini's theorem,

$$\begin{aligned}
\mathbb{E}[xy] &= \int_{\mathbb{R}^2} (xy) \mathbb{P}_{(x,y)}(dxdy) \\
&= \cdots \\
&= \mathbb{E}(x) \cdot \mathbb{E}(y)
\end{aligned}$$

□

Proposition 3.1.6 (Independence of Generated σ-Algebras)**.** *For a family of collections of sets $(\mathcal{C}i)_{i\in I}$, if they are mutually independent and closed under intersections, then $(\sigma(\mathcal{C}i))_{i\in I}$ are mutually independent.*

Proof. Consider $J = \{i_1, \ldots, i_n\} \subset\subset I$, and without loss of generality, assume that$\#(J) > 1$. Then,

$$\mathbb{P}\left(A_{i_1} \cap \ldots \cap A_{in}\right) = \prod_{k=1}^{n} \mathbb{P}\left(A_{ik}\right) \tag{3.1.2}$$

Where $A_{i_k} \in \mathcal{C}_{i_k}, k = 1, \cdots, n$. Fixing $A_{i_2}, \cdots, A_{i_n}$, define $\mathcal{D} := \{A_{i_1} \in \mathcal{A} : 3.1.2 \text{ holds}\}$. It is easy to verify that $\mathcal{D}$ is a Dynkin system.
(i) $\Omega \in \mathcal{D}$
(ii) $A \in \mathcal{D}$, then $A^c \in \mathcal{D}$

$$\begin{aligned}
\mathbb{P}\left(A^C \cap \left(\bigcap_{k=2}^{n} A_{ik}\right)\right) &= \mathbb{P}\left((\Omega \backslash A) \cap \left(\bigcap_{k=2}^{n} A_{ik}\right)\right) \\
&= \mathbb{P}\left(\Omega \cap \left(\bigcap_{k=2}^{n} A_{ik}\right)\right) - \mathbb{P}\left(A \cap \left(\bigcap_{k=2}^{n} A_{ik}\right)\right) \\
&= \left(\mathbb{P}(\Omega) - \mathbb{P}(A)\right) \cdot \prod_{k=2}^{n} \mathbb{P}\left(A_{ik}\right) \\
&= \mathbb{P}\left(A^c\right) \cdot \prod_{k=2}^{n} \mathbb{P}\left(A_{ik}\right).
\end{aligned}$$

(iii) For any $A_{1_1}, A_{2_1}, \ldots \in \mathcal{D}$ and $A_{i_1} \bigcap A_{j_1} = \varnothing, \forall i \neq j$, we have

$$\begin{aligned}
\mathbb{P}\left(\left(\bigcup_{j=1}^{\infty} A_{j_1}\right) \cap \left(\bigcap_{k=2}^{n} A_{i_k}\right)\right) &= \sum_{j=1}^{\infty} \mathbb{P}\left(A_j \cap \left(\bigcap_{k=2}^{\infty} A_{i_k}\right)\right) \\
&= \sum_{j=1}^{\infty} \mathbb{P}\left(A_{j_1}\right) \cdot \prod_{k=2}^{n} \mathbb{P}\left(A_{i_k}\right) \\
&= \mathbb{P}\left(\bigcup_{j=1}^{\infty} A_{j_1}\right) \cdot \prod_{k=2}^{n} \mathbb{P}\left(A_{i_k}\right).
\end{aligned}$$

Since $\mathcal{C}_{i_1} \subset \mathcal{D}$ is closed under intersections, and $\mathcal{C}_{i_1} \subset \mathcal{D}$, it follows that $\sigma(\mathcal{C}_{i_1}) \subset \mathcal{D}$ Therefore, equation 3.1.2 holds for $A_{i_1} \in \sigma(\mathcal{C}_i), A_{i_2} \in \mathcal{C}_{i_2}, \cdots, A_{i_n} \in \mathcal{C}_{i_n}$. Repeating the above argument for $k = 2, \cdots, n$, the proposition is proved. □

Corollary 3.1.7 (Independence of Indicator Functions)**.** *A family of events $(A_i)_{i \in I}$ is mutually independent if and only if the corresponding family of random variables formed by their indicator functions $(\mathbb{1}_{A_i})_{i \in I}$ is mutually independent. In particular, for $J \subset\subset I, B_j \in \{A_j, A_j^c\}, j \in J$, $\mathbb{P}(\bigcap_{j \in J} B_j) = \prod_{j \in J} \mathbb{P}(B_j)$.*

Proof. For $i \in I$, let $\mathcal{C}_i = \{A_i\}$, then

$$\sigma\left(\mathbb{1}_{A_i}\right) = \{\phi, A_i, A_i^c, \Omega\} = \sigma\left(C_i\right).$$

Since $\mathcal{C}_i$ is closed under intersections, the proposition follows from Proposition **??**. □

Lemma 3.1.8 (Borel-Cantelli Lemma)**.** *(i)Suppose $A_1, A_2, \ldots \in \mathcal{A}$ is a sequence of events in a probability space. If the sum of their probabilities is finite, i.e., $\sum_{n=1}^{\infty} \mathbb{P}\left(A_n\right) < \infty$, then $\mathbb{P}\left(\limsup_{n \to \infty} A_n\right) = 0$.*
(ii)If $A_1, A_2, \ldots$ are independent and $\sum_{n=1}^{\infty} \mathbb{P}\left(A_n\right) = \infty$, then $\mathbb{P}\left(\limsup_{n \to \infty} A_n\right) = 1$.

Proof. (i) By the continuity of the probability measure $\mathbb{P}$,

$$\begin{aligned}\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) &= \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{m=n}^{\infty} A_m\right)\\ &\leqslant \lim_{n\to\infty} \sum_{m=n}^{\infty} \mathbb{P}(A_m)\\ &= 0.\end{aligned}$$

(ii)Applying the inequality$1-x \leqslant e^{-x}, \forall x \geqslant 0$ (since$1-e^x = e^0 - e^x$, and by the Lagrange mean value theorem, there exists $\xi \in [-x, 0],$ s.t. $e^0 - e^x = e^{\xi}\cdot x \leqslant x$). Then, by the continuity of the probability measure $\mathbb{P}$, we have:

$$\begin{aligned}\mathbb{P}\left(\left(\limsup_{n\to\infty} A_n\right)^c\right) &= \mathbb{P}\left(\bigcup_{n=1}^{\infty}\bigcap_{m=n}^{\infty} A_m^c\right)\\ &= \lim_{n\to\infty} \mathbb{P}\left(\bigcap_{m=n}^{\infty} A_m^c\right).\end{aligned}$$

Then, using the independence of (A_n), we have

$$\begin{aligned}\mathbb{P}\left(\bigcap_{m=n}^{\infty} A_m^c\right) &= \lim_{N\to\infty} \mathbb{P}\left(\bigcap_{m=n}^{N} A_m^c\right)\\ &= \lim_{N\to\infty} \prod_{m=n}^{N} \mathbb{P}(A_m^c)\\ &= \lim_{N\to\infty} \prod_{m=n}^{N} (1-\mathbb{P}(A_m))\\ &\leqslant \lim_{N\to\infty} \prod_{m=n}^{N} \exp(-\mathbb{P}(A_m))\\ &= \lim_{N\to\infty} \exp\left(-\sum_{m=n}^{N} \mathbb{P}(A_m)\right)\\ &= 0.\end{aligned}$$

Thus, $\mathbb{P}\left(\left(\lim_{n\to\infty}\sup A_n\right)^c\right) = 0,\ \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 1$ □

Exercises

1. Let $(X_n : n \in \mathbb{N})$ be independent exponentially distributed random variables, i.e.,

   $$P(X_n > x) = e^{-x}, \quad x \geqslant 0.$$

   Then for $\alpha > 0$

   $$P(X_n > \alpha \log n) = n^{-\alpha}.$$

   Let $A_n := \{X_n > \alpha \log n\}$, please prove:

   $$P(\limsup_{n\to\infty} A_n) = \begin{cases} 0 & \text{if } \alpha > 1,\\ 1 & \text{if } \alpha \leqslant 1.\end{cases}$$

2. Let $(\mathcal{F}_i)_{i\in I}$ be a family of independent σ-algebras. Furthermore, let $\mathcal{I}$ be a partition of I, that is, $\mathcal{I} = \{I_k, k \in K\}$ and $\biguplus_{k\in K} I_k = I$, where the I_k are pairwise disjoint and their union is I. Then $(\sigma(\mathcal{F}_i : i \in I_k))_{k\in K}$ is also an independent system.
3. Let $A_1, A_2, \cdots$ be a sequence of independent events, and for all $n \in \mathbb{N}$, there is $P(A_n) < 1$. Prove: if $P(\cup_{n=1}^{\infty} A_n) = 1$, then $P(\limsup_{n\to\infty} A_n) = 1$.
4. Let $X_1, X_2, \cdots$ be independent random variables. Prove: $\sup_{n\in\mathbb{N}} X_n < \infty$ holds almost surely if and only if for all $c \in \mathbb{R}$, we have $\sum_{n=1}^{\infty} P(X_n > c) < \infty$.

3.2 Existence of Sequences of Independent Random Variables

Definition 3.2.1. Let $(\Omega_i, \mathcal{A}_i)$ be measurable spaces for each $i \in I$.
Define $\times_{i\in I}\Omega_i := \{(\omega_i)_{i\in I} : \omega_i \in \Omega_i,\ \forall i \in I\}$as the Cartesian product of the Ω_i.
The mapping$\pi_j : \times_{i\in I}\Omega_i \longrightarrow \Omega_j$is called the projection of $\boldsymbol{\omega}$ onto the component Ω_j, defined by$\pi_j(\boldsymbol{\omega}) = \omega_j,\ \forall \boldsymbol{\omega} \in \times_{i\in I}\Omega_i$.

Definition 3.2.2 (Product σ-Algebra)**.** On$\times_{i\in I}\Omega_i$, we define the product σ-algebra as

$$\bigotimes_{i\in I}\mathcal{A}_i := \sigma\{\underbrace{A_i \times \left(\bigtimes_{j\neq i}\Omega_j\right)}_{n\text{-dimensional cylinder sets}} : i \in I,\ A_i \in \mathcal{A}_i\}.$$

It is the smallest σ-algebra on $\times_{i\in I}\Omega_i$ that makes all projection operators measurable.

Theorem 3.2.3 (Existence of a Bernoulli Sequence)**.** *There exists a sequence of independent and identically distributed random variables such that each random variable follows a Bernoulli distribution with parameter* $\frac{1}{2}$*.*

Proof. We first consider the space $0, 1$ and define the measure $\mu := \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$. Consider $\mathcal{P}(\{0,1\})$, the σ-algebra consisting of all subsets of $\{0,1\}$ Let $\Omega := \{0,1\}^{\mathbb{N}}$, and define $\gamma_n : \Omega \longrightarrow \{0,1\}^n$ as the projection of Ω onto the first n components. Define $Q_n := \mu^{\otimes n}$ as the product measure of μ on $\mathcal{A}_n := \mathcal{P}(\{0,1\}^n)$. Define a class of sets closed under intersection and complementation, which is $\varepsilon := \bigcup_{n\in\mathbb{N}} \gamma_n^{-1}(\mathcal{A}_n)$. This class is the generator of the the product σ-algebra $\mathcal{A} := \bigotimes_{n\in\mathbb{N}} P(\{0,1\}) = P(\{0,1\}) \otimes \cdots \otimes P(\{0,1\})$. Define $Q\left(\gamma_n^{-1}(B)\right) := Q_n(B)$, which is well-defined. If for $n \leqslant m$ we have $\gamma_n^{-1}(B) = \gamma_m^{-1}(C)$, then $C = B \times (\bigtimes_{n<k\leqslant m} \{0,1\})$ and $Q_n(B) = Q_m(C)$.Due to the independence implied in the above representation, we obtain $Q(E \bigcup F) = Q(E) + Q(F)$ for all disjoint $E, F \in \varepsilon$.

Construct a probability measure $\mathbb{P}$ on $\mathcal{A}$ such that it agrees with Q on the class ε. From this, define the projections

$$X_n := \pi_n : (\Omega, \mathcal{A}) \longrightarrow (\{0,1\}, \mathcal{P}(\{0,1\})),\ n \in \mathbb{N}$$

and for$A_n \subset \{0,1\}$, we have the inequality

$$\begin{aligned}\mathbb{P}\left(\bigcap_{n=1}^{m}\{X_n \in A_n\}\right) &= Q\left(\bigcap_{n=1}^{m}\{X_n \in A\}\right)\\ &= Q_m\left(A_1 \times \ldots \times A_m\right)\\ &= \frac{1}{2^m}\#\left(A_1 \times \ldots \times A_m\right)\\ &= \prod_{n=1}^{m}\mathbb{P}\left(X_n \in A_n\right)\end{aligned}$$

This shows that $(X_n)_n$ are independent, and $(X_n) \sim \mathrm{Ber}_{1/2},\ n \in \mathbb{N}$.

Next, we prove that Q is σ-additive on ε (and from this we can obtain countable sub-σ-additivity.), that is, we prove that for any $E_m \in \varepsilon,\ E := \biguplus_{m=1}^{\infty} E_m \in \varepsilon$, we have $Q\left(\biguplus_{m=1}^{\infty} E_m\right) = \sum_{m=1}^{\infty} Q\left(E_m\right)$. The reason is that E must consist of only finitely many of the E_m, and the remaining part must be $\varnothing$, so σ-additivity can be derived from finite additivity. We prove this fact by contradiction.

Assume that for every $m \in \mathbb{N}$, there exists a sequence $X_m = (X_{m,k})_{k\in\mathbb{N}} \in E\backslash\bigcup_{j=1}^{m} E_j$. Since $\{X_{m,k},\ m,k \in \mathbb{N}\} \subset \{0,1\}$, by induction we can show that there exist a sequence of sets $M_{k+1} \subset M_k \subset \mathbb{N}$ and $y_k \in \{0,1\}$, such that $(X_{m,1,\ldots,}, X_{m,k}) = (y_1 \ldots, y_k)\,,\ m \in M_k$. If $E \in \gamma_n^{-1}\left(\mathcal{A}_n\right)$, this means that whether a sequence is in E depends only on its first n coordinates. From this we can deduce that $y = (y_k)_{k\in\mathbb{N}} \in E$, because $\gamma_n(y) = \gamma_n\left(X_m\right)\,,\ \forall m \in M_n$. If $y \in E_m \in \gamma_k^{-1}\left(\mathcal{A}_k\right)\,,\ m,k \in \mathbb{N}$ then there exists $p \in M_k,\ p \geqslant m$, since $\gamma_k(y) = \gamma_k\left(X_p\right)$, it must be that $X_p \in E_m \subset \bigcup_{j=1}^{p} E_j$, which contradicts the assumption that $X_p \in E\backslash\bigcup_{j=1}^{p} E_j$.

For $\forall A \subset \Omega$, define $\mathbb{P}(A) := \inf\left\{\sum_{m=1}^{\infty} Q\left(E_m\right) : E_m \in \varepsilon, A \subset \bigcup_{m=1}^{\infty} E_m\right\}$. We will prove that $\mathbb{P}$ is a probability measure on $\mathcal{A} = \theta(\varepsilon)$ and that $\mathbb{P}\left(E\right) = Q(E), \quad \forall E \in \varepsilon$. Since $\varnothing = \bigcup_{n\in\mathbb{N}} \varnothing_n$, we have $\mathbb{P}(\varnothing) = 0$. Moreover, from the countable subadditivity of Q, we have $1 = Q(\Omega) \leqslant \sum_{m=1}^{\infty} Q\left(E_m\right), \quad \forall\left(E_m\right) \subset \varepsilon$, and $\bigcup_{m\in\mathbb{N}} E_m = \Omega$, therefore $\mathbb{P}(\Omega) \geqslant 1^*$. And since $\Omega = \Omega\bigcup(\bigcup_{n=2}^{\infty} \varnothing, \mathbb{P}(\Omega) \leqslant 1$, we obtain $\mathbb{P}(\Omega) = 1$. We now prove that $\mathbb{P}$ is countably subadditive. Suppose for $A_n \subset \Omega, \varepsilon > 0$ we have $A_n \subset \bigcup_{m\in\mathbb{N}} E_{n,m}$ and $\mathbb{P}\left(A_n\right) + \varepsilon/2^n \geqslant \sum_{m=1}^{\infty} Q\left(E_{n,m}\right)$. Since $\bigcup_{n\in\mathbb{N}} A_n \subset \bigcup_{m,n\in\mathbb{N}} E_{n,m}$, we obtain:

$$\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leqslant \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} Q(E_{n,m}) \leqslant \varepsilon + \sum_{n=1}^{\infty}\mathbb{P}(A_n).$$

Therefore,

$$\mathcal{D} := \{A \subset \Omega : \mathbb{P}(T) = \mathbb{P}(T \cap A) + \mathbb{P}(T\backslash A), \quad \forall T \subset \Omega\}$$

is a Dynkin class, and $\mathbb{P}$ is σ-additive on $\mathcal{D}$. (Left as an exercise.)

Next, we prove that $\varepsilon \subset \mathcal{D}$. If$E \in \varepsilon, T \subset \Omega$ and $\varepsilon > 0$, then there exist $E_m \in \varepsilon$ such that $T \subset \bigcup_{m\in\mathbb{N}} E_m$ and

$$\begin{aligned}\mathbb{P}(T) + \varepsilon &\geqslant \sum_{m=1}^{\infty} Q\left(E_m\right)\\ &= \sum_{m=1}^{\infty} Q\left(E_m \cap E\right) + \sum_{m=n}^{\infty} Q\left(E_m\backslash E\right)\\ &\geqslant \mathbb{P}(T \cap E) + \mathbb{P}(T\backslash E).\end{aligned}$$

Let$\varepsilon \to 0$, we obtain that

$$\begin{aligned}\mathbb{P}(T \cap E) + \mathbb{P}(T \backslash E) &\leqslant \mathbb{P}(T) \\ &= \mathbb{P}((T \cap E) \cup (T \backslash E)) \\ &\leqslant \mathbb{P}(T \cap E) + \mathbb{P}(T \backslash E).\end{aligned}$$

From this, we can conclude that $\boldsymbol{\varepsilon} \subset \mathcal{D}$, and since it's a Dynkin class, we have $\mathcal{A} = \sigma(\boldsymbol{\varepsilon}) \subset \mathcal{D}$. Therefore, we have proven that $\mathbb{P}$ is a probability measure on $\mathcal{A}$. Next, we prove that $\mathbb{P}$ agrees with Q on $\boldsymbol{\varepsilon}$.

Choose $E \in \boldsymbol{\varepsilon}$, then $\mathbb{P}(E) \leqslant Q(E)$. Since $E \subset E \cup \left(\bigcup_{n \geqslant 2} \varnothing\right)$, in particular we also have $\mathbb{P}\left(E^C\right) \leqslant Q\left(E^c\right)$. Therefore $\mathbb{P}(E) = 1 - \mathbb{P}\left(E^c\right) \geqslant 1 - Q\left(E^c\right) = Q(E)$ □

Theorem 3.2.4 (Uniform Distribution on $[0,1]$)**.** *There exists a sequence of independent random variables $(X_n)_{n \geqslant 1}$ such that for any $I \subset [0,1]$, the length of I is equal to$\mathbb{P}\left(X_n \in I\right)$.*

Proof. According to Theorem 3.2.3, there exists a family of independent random variables($Y_{n,m})_{(n,m) \in \mathbb{N}^2}$, where each $Y_{n,m} \sim \text{Ber}_{1/2}$. Suppose $X_n := \sum_{m=1}^{\infty} 2^{-m} Y_{n,m}$, then X_n is a random variable with $X_n : (\Omega, \mathcal{A}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. And since $\sigma\left(X_n\right) \subset \sigma\left(Y_{n,m} : m \in \mathbb{N}\right)$, $\left(X_n\right)_n$ are independent random variables (left as an exercise). The sequence $(Y_{n,m})_m$ can be viewed as the binary digit expansion of X_n. To obtain the uniqueness of this binary representation, we observe the following fact.

$$A_n := \limsup \{Y_{n,m} = 0\} = \{Y_{n,m} = 0, \text{for finitely many } m \in \mathbb{N}\}$$

Since $\mathbb{P}\left(y_{n,m} = 0\right) = 1/2$ by the Borel-Cantelli lemma, we have $\mathbb{P}\left(A_n\right) = 1$.
Let $J := \left[\sum_{i=1}^{j} 2^{-i} \delta_i, 2^{-j} + \sum_{i=1}^{j} 2^{-i} \delta_i\right)$, $\delta_i \in \{0,1\}$, where J is a dyadic interval with length $l(j) = 2^{-j}$. Then we have

$$\begin{aligned}\mathbb{P}\left(X_n \in J\right) &= \mathbb{P}\left(A_n \bigcap \{X_n \in J\}\right) \\ &= \mathbb{P}\left(A_n \bigcap \{y_{n,1} = \delta_1, \ldots; y_{n,j} = \delta_j\}\right) \\ &= \mathbb{P}\left(\bigcap_{i=1}^{j} \{y_{n,i} = \delta_i\}\right) \\ &= \prod_{i=1}^{j} \mathbb{P}\left(y_{n,i} = \delta_i\right) \\ &= 2^{-j} \\ &= l(j).\end{aligned}$$

Since for every interval $I = [a,b]$, if its endpoints lie in the set $\left\{j \cdot 2^{-k} : j \in \mathbb{N}_0, k \in \mathbb{N}\right\}$, then it must be a union of the aforementioned dyadic intervals, so $\mathbb{P}\left(X_n \in I\right) = l(I)$.
Finally, consider any interval K and $\varepsilon > 0$ such that $I_0 \subset K \subset I_1$ and $l\left(I_1\right) - l\left(I_0\right) < \varepsilon$, I_0, I_1 are intervals as described above. Then, by considering cases based on the absolute value sign, we obtain

$$\left|\mathbb{P}\left(X_n \in K\right) - l(K)\right| < \varepsilon$$

□

Theorem 3.2.5 (Existence of a Sequence of Independent Random Variables)**.** *For any sequence of distributions, $\mu_n$are probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then there exists a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and a sequence of random variables $(X_n)_{n\in\mathbb{N}}$ defined on it such that each random variable $X_n : (\Omega, \mathcal{A}) \longrightarrow (\mathbb{R}, B(\mathbb{R}))$ has distribution $X_n \sim \mu_n$.*

Proof. Let $F_n(x) := \mu_n((-\infty, x])$. According to Theorem 3.2.4, there exists a family of independent random variables $(Z_n)_{n\in\mathbb{N}}$ such that $Z_n \sim U_{[0,1]}$. Define $Y_n = g(Z_n)$,$g : \mathbb{R} \to \mathbb{R}$ where

$$g(x) := \begin{cases} x, & x \in (0, 1); \\ \frac{1}{2}, & \text{otherwise.} \end{cases}$$

Then $Y_n \sim U_{[0,1]}$ and $y_n : \Omega \to (0, 1), n = 1, 2, \cdots$. Now we define $X_n = F_n^{-1}(Y_n)$ (by Property 2.0.4), so that the random variables $X_n, n = 1, 2, \cdots$, are independent and satisfy $X_n \sim \mu_n$.

□

Theorem 3.2.6 (Existence of the Lebesgue Measure)**.** *There exists a measure λ on $\mathcal{B}(\mathbb{R})$ such that $\lambda(I) = l(I),\ \forall I \subset R$.*

Proof. According to Theorem 3.2.4, there exists a random variable Y on $(\Omega, \mathcal{A}, \mathbb{P})$ that follows the distribution of $U_{[0,1]}$, satisfying $\mathbb{P}_Y(I) = l(I),\ \forall I \subset [0, 1]$. For an y $z \in \mathbb{Z}$, define $W_z := (z, z + 1]$. Then $\mathbb{P}_{Y+z}(I) = l(I),\ \forall I \subset W_z$.
For any $A \in \mathcal{B}(\mathbb{R})$, define $\lambda(A) := \sum_{z\in\mathbb{Z}} \mathbb{P}_{Y+z}(A \bigcap W_z)$. Since $\mathbb{P}_{Y+z}$ is a measure on $\mathcal{B}(\mathbb{R})$, it follows that $\lambda$is also a measure. Because $\{W_z : z \in \mathbb{Z}\}$ is a partition of $\mathbb{R}$, for any interval I, by the σ-additivity of λ and the additivity of length, we have

$$\begin{aligned} \lambda(I) &= \sum_{z\in\mathbb{Z}} \mathbb{P}_{y+z}\left(I \bigcap W_z\right) \\ &= \sum_{z\in\mathbb{Z}} \mathbb{P}_{y+z}(I \bigcap (z, z + 1]) \\ &= \sum_{z\in\mathbb{Z}} l(I \bigcap (z, z + 1]) \\ &= l(I). \end{aligned}$$

□

Exercises

1. On the power set 2^Ω of Ω, a mapping $\eta : 2^\Omega \to [0, \infty]$ is called an outer measure if it satisfies the following conditions: (i) $\eta(\varnothing) = 0$, (ii) σ-subadditivity: for all $A, A_1, A_2, \cdots \in \Omega$ with $A \subset \cup_{n=1}^{\infty} A_n$, we have $\eta(A) \leqslant \sum_{n=1}^{\infty} \eta(A_n)$.

   If for all $C \subset \Omega$ it holds that

   $$\eta(C \cap A) + \eta(C\backslash A) = \eta(C)$$

   then the subset $A \subset \Omega$ is called η-measurable.

   Suppose η is an outer measure defined on Ω. Prove that the system $\mathcal{A}$ of all η-measurable sets is a Dynkin system, and that the restriction of η to $\mathcal{A}$ is σ-additive.

Chapter 4

Convergence of Sequences of Random Variables: Almost Surely, In Probability, and L^p-Convergence

4.1 L^p-Spaces

Definition 4.1.1 ($\mathcal{L}^p$-Space)**.** For $0 < p \leqslant \infty$, we define

$$\mathcal{L}^p := \mathcal{L}^p(\Omega, \mathcal{A}, \mathbb{P}) := \{X : \Omega \to \mathbb{R} \text{is a random variable and} \|X\|_p < \infty\}$$

where$\|X\|_p := (\mathbb{E}[|X|]^p)^{1/p}, 0 < p < \infty$,$\|X\|_\infty := \inf\{k : \mathbb{P}(|X| > k) = 0\}$

Remark 4.1.2. (i)We have

$$\begin{aligned}\|X\|_p = 0 &\Leftrightarrow \int_\Omega |X|^p d\mathbb{P} = 0. \\ &\Leftrightarrow X = 0 \quad \text{almost surely.}\end{aligned}$$

(ii) $\|\alpha \cdot X\|_p = |\alpha| \|X\|_p \quad \forall \alpha \in \mathbb{R}$.

(iii) From the estimate

$$|x + y|^p \leqslant (|x| + |y|)^p \leqslant (2|x|)^p + (2|y|)^p$$

we obtain

$$\int_\Omega |X + Y|^p d\mathbb{P} \leqslant 2^p \int_\Omega |X|^p d\mathbb{P} + 2^p \int_\Omega |Y|^p d\mathbb{P}$$

Therefore, if $X, Y \in \mathcal{L}^p$, then $X + Y \in \mathcal{L}^p$.

Thus, $\mathcal{L}^p$ is a vector space ($\alpha X + \beta Y \in \mathcal{L}^p$ For $\alpha, \beta \in \mathbb{R}, X, Y \in \mathcal{L}^p$).

Proposition 4.1.3 (Minkowski Inequality)**.** *If $X, Y \in \mathcal{L}^p$ $1 \leqslant p \leqslant \infty$, then*

$$\|X + Y\|_p \leqslant \|X\|_p + \|Y\|_p$$

Proof. For $p = 1$, the result directly follows from

$$|x + y| \leqslant |x| + |y|$$

For the case $p = \infty$, it is similarly simple. Therefore, assume $1 < p < \infty$. Let $q := p/(p-1) > 1$, then $1/p + 1/q = 1$. We have

$$\int_\Omega |X+Y|^p d\mathbb{P} \leqslant \int_\Omega |X| \cdot |X+Y|^{p-1} d\mathbb{P} + \int_\Omega |Y||X+Y|^{p-1} d\mathbb{P}$$

Using Hölder's inequality, we obtain

$$\int_\Omega |X+Y|^p d\mathbb{P} \leqslant \left[\left(\int_\Omega |X|^p d\mathbb{P} \right)^{1/p} + \left(\int_\Omega |Y|^p d\mathbb{P} \right)^{1/p} \right] \cdot \left(\int_\Omega |X+Y|^{(p-1)\cdot q} d\mathbb{P} \right)^{1/q}$$

Since $(p-1) \cdot q = p$ and $1 - 1/q = 1/p$ we conclude the result. The special case where $\int_\Omega |X+Y|^p d\mathbb{P} = 0$ is obvious. □

Definition 4.1.4. (p-th Mean Convergence) If a sequence $X_1, X_2, \ldots \in \mathcal{L}^p$ converges in $\mathcal{L}^p$ (or in p-th mean) to $X \in \mathcal{L}^p$, that is

$$\|X_n - X\|_p \xrightarrow{n\to\infty} 0$$

We can also write this as $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p} X$.

Proposition 4.1.5. *($\mathcal{L}^p$ Completeness) If $p \geqslant 1$ and $X_1, X_2, \ldots$ is a Cauchy sequence in $\mathcal{L}^p$, that is, $\lim_{n\to\infty} \|X_n - X_m\|_p = 0$, then there exists an $X \in \mathcal{L}^p$ such that $\|X_n - X\|_p \xrightarrow{n\to\infty} 0$.*

Proof. For the case $p = \infty$, the result is obvious. Assume $1 \leqslant p < \infty$. Let $\varepsilon_1, \varepsilon_2, \ldots$ be a summable sequence, for example, $\varepsilon_n := 2^{-n}$. Since $X_1, X_2, \ldots$ is a Cauchy sequence, for each k, there exists an index n_k such that

$$\|X_{n_k} - X_n\|_p \leqslant \varepsilon_k \quad \forall n \geqslant n_k$$

In particular, we have

$$\sum_{k=1}^{\infty} \|X_{n_{k+1}} - X_{n_k}\|_p \leqslant \sum_{k=1}^{\infty} \varepsilon_k < \infty$$

By the Monotone Convergence Theorem and the Minkowski inequality, we obtain

$$\begin{aligned} & \left\| \textstyle\sum_{k=1}^{\infty} |X_{n_{k+1}} - X_{n_k}| \right\|_p \leqslant \textstyle\sum_{k=1}^{\infty} \|X_{n_{k+1}} - X_{n_k}\|_p < \infty \\ \Leftrightarrow \quad & \lim_{N\to\infty} \left\| \textstyle\sum_{k=1}^{N} |X_{n_{k+1}} - X_{n_k}| \right\|_p \leqslant \textstyle\sum_{k=1}^{\infty} \|X_{n_{k+1}} - X_{n_k}\|_p < \infty \\ \Leftrightarrow \quad & \lim_{N\to\infty} \textstyle\sum_{k=1}^{N} \|X_{n_{k+1}} - X_{n_k}\|_p \leqslant \textstyle\sum_{k=1}^{\infty} \|X_{n_{k+1}} - X_{n_k}\|_p < \infty \end{aligned}$$

Thus, in particular, we have $\sum_{k=1}^{\infty} |X_{n_{k+1}} - X_{n_k}| < \infty$ for almost every ω, so $X_{n_1}(\omega), X_{n_2}(\omega), \ldots$ is a Cauchy sequence of real numbers for almost every ω. Therefore, there exists a random variable X such that

$$X_{n_k} \xrightarrow{k\to\infty} X \quad \text{a.e.}$$

By Fatou's Lemma, we have

$$\begin{aligned}\|X_n - X\|_p &= \left\| X_n - \lim_{k\to\infty} X_{n_k} \right\|_p \\ &\leqslant \liminf_{k\to\infty} \|X_{n_k} - X_n\|_p \\ &\leqslant \sup_{m\geqslant n} \|X_m - X_n\|_p \xrightarrow{n\to\infty} 0\end{aligned}$$

That is, $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p} X$. □

4.2 Almost Sure Convergence and Convergence in Probability

Definition 4.2.1. Let $X, X_1, X_2, \ldots$ be random variables.

(i) If $\mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1$, we say that the sequence $X_1, X_2, \ldots$ converges almost surely to X, and we write

$$X_n \xrightarrow[n\to\infty]{f.s.} X.$$

(ii) If $\lim_{n\to\infty} \mathbb{P}\left(|X_n - X| > \varepsilon\right) = 0 \quad \forall \varepsilon > 0$, we say that the sequence $X_1, X_2, \ldots$ converges in probability (or randomly) to X, and we write

$$X_n \xrightarrow[n\to\infty]{p} X.$$

Theorem 4.2.2. *Let $X, X_1, X_2, \ldots$ be random variables. Then, the following statements are equivalent:*

(i) $X_n \xrightarrow[n\to\infty]{p} X$

(ii) Every subsequence of $X_1, X_2, \ldots$ contains a subsequence that converges almost surely to X.

Proof. First, assume (i) holds. Then, for every subsequence, we can find a subsequence $1 \leqslant n_1 < n_2 < \ldots$ such that

$$\mathbb{P}\left(|X_{n_k} - X| > 2^{-k}\right) \leqslant 2^{-k}.$$

By the Borel-Cantelli Lemma, we have

$$\mathbb{P}\left(|X_{n_k} - X| > 2^{-k} \text{ for infinitely many } k\right) = 0.$$

This means that $X_{n_1}, X_{n_2}, \ldots$ converges almost surely to X.

Next, assume (ii) holds, and let $1 \leqslant n_1 < n_2 < \ldots$ be a subsequence satisfying condition (ii). For any $\varepsilon > 0$, we have

$$\mathbb{1}_{\{|X_{n_k} - X| > \varepsilon\}} \xrightarrow{\text{a.s.}} 0.$$

By the Dominated Convergence Theorem, we have

$$\mathbb{P}\left(|X_{n_k} - X| > \varepsilon\right) = \int_\Omega \mathbb{1}_{\{|X_{n_k} - X| > \varepsilon\}} d\mathbb{P} \xrightarrow{k\to\infty} 0.$$

Thus, any subsequence of $\mathbb{P}(|X_n - x| > \varepsilon)$ contains a subsequence that converges to 0. Therefore, the original sequence $\mathbb{P}(|X_n - X| > \varepsilon)$ converges to 0. Hence, (i) holds. □

Corollary 4.2.3. *Let $X, X_1, X_2, \ldots$ be random variables. Then, we have the following:*

(i) If $X_n \xrightarrow[n\to\infty]{p} X$, then there exists a subsequence X_{n_k} such that $X_{n_k} \xrightarrow[k\to\infty]{a.s} X$.

(ii) If $X_n \xrightarrow[n\to\infty]{a.s} X$, then $X_n \xrightarrow[n\to\infty]{p} X$.

Exercise

1. Prove: If $X_n \to X$ converges in probability, and $Y_n \to Y$ converges in probability, then $X_n \pm Y_n \to X \pm Y$ converges in probability, and $X_n Y_n \to XY$ converges in probability.

4.3 Convergence in Probability and L^p-Convergence

Theorem 4.3.1. *Let X and $X_1, X_2, \ldots$ be elements of $\mathcal{L}^p$, where $1 \leqslant p < \infty$. Then the following statements are equivalent:*

(i) $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p} X$

(ii) $X_n \xrightarrow[n\to\infty]{p} X$ and $\|X_n\|_p \xrightarrow{n\to\infty} \|X\|_p$.

Proof. (i) ⇒ (ii): By Chebyshev's inequality, we immediately have

$$\mathbb{P}(|X_n - X| > \varepsilon) \leqslant 1/\varepsilon^p \int_\Omega |X_n - X|^p \, d\mathbb{P}$$

This proves convergence in probability. By Minkowski's inequality, we have

$$\left| \|X_n\|_p - \|X\|_p \right| \leqslant \|X_n - X\|_p \xrightarrow{n\to\infty} 0$$

This proves the convergence $\|X_n\|_p \xrightarrow{n\to\infty} \|X\|_p$.

(ii) ⇒ (i): Since $X_n \xrightarrow{p} X$, for every subsequence, we can find a subsequence $1 \leqslant n_1 < n_2 < \ldots$ such that $X_{n_k} \xrightarrow[k\to 0]{a.s.} X$. Applying Fatou's lemma to

$$2^p (|X|^p + |X_{n_k}|^p) - |X_{n_k} - X|^p \geqslant 0,$$

we obtain

$$2^p \int_\Omega 2 \cdot |X|^p d\mathbb{P} \leqslant \liminf_{k\to\infty} \left(2^p \int_\Omega |X|^p \, d\mathbb{P} + 2^p \int_\Omega |X_{n_k}|^p \, d\mathbb{P} - \int_\Omega |X_{n_k} - X|^p \, d\mathbb{P} \right)$$

By the assumption, both sides contain the term $2^p \int_\Omega |X|^p d\mathbb{P}$twice. Since it is finite, we have

$$\limsup_{k\to\infty} \int_\Omega |X_{n_k} - X|^p \, d\mathbb{P} \leqslant 0.$$

Overall, every subsequence has a subsequence such that $\|X_n - X\|_p$ converges to 0. This is equivalent to $\|X_n - X\|_p \longrightarrow 0$. □

Definition 4.3.2. Let $1 \leqslant p < \infty$. If a family of random variables $(X_i)_{i\in I}$ satisfies

$$\lim_{M\to\infty}\left(\sup_{i\in I}\int_{\{|X_i|>M\}}|X_i|^p\,d\mathbb{P}\right)=0,$$

we say that the family of random variables is uniformly p-integrable.

Theorem 4.3.3 (Vitali)**.** *Let $X, X_1, X_2, \ldots \in \mathcal{L}^p$, where $1 \leqslant p < \infty$. Then the following statements are equivalent:*

(i) $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p} X$

(ii) $X_n \xrightarrow[n\to\infty]{p} X$ *and* $(X_n)_{n\in\mathbb{N}}$ *is uniformly p-integrable.*

Proof. $(i) \Rightarrow (ii)$ Let $1 \leqslant n_1 < n_2 < \ldots$ be a subsequence chosen as in the proof of Theorem 4.3.1such that $X_{n_k} \xrightarrow[k\longrightarrow\infty]{a.s.} X$. Then

$$|X_{n_k}|^p \cdot \mathbb{1}_{\{|X_{n_k}|<M\}} \to |X|^p \mathbb{1}_{\{|X|<M\}} \quad \text{a.s.}$$

By the Dominated Convergence Theorem (first along the subsequence and then extending to the entire sequence), we have

$$\int_{\{|X_n|<M\}} |\,X_n|^p d\mathbb{P} \longrightarrow \int_{\{|X|<M\}} |X|^p d\mathbb{P}.$$

From the assumption of (i), we have

$$\int_{\{|X_n|\geqslant M\}} |X_n|^p\,d\mathbb{P} \longrightarrow \int_{\{|X|\geqslant M\}} |X|^p d\mathbb{P}. \tag{4.3.1}$$

Given $\varepsilon > 0$, since

$$\lim_{M\to\infty}\int_{\{|X|\geqslant M\}}|X|^p d\mathbb{P}=0,$$

we can find an M_0 such that

$$\int_{\{|X|\geqslant M_0\}}|X|^p d\mathbb{P}<\varepsilon/2.$$

From (4.3.1), there exists an $N_0 \in \mathbb{N}$ such that

$$\int_{\{|X_n|\geqslant M_0\}}|X_n|^p\,d\mathbb{P}<\varepsilon \text{ for } n>N_0.$$

For $n = 1, 2, \ldots, N_0$, we have

$$\lim_{M\to\infty}\int_{\{|x_n|\geqslant M\}}|X_n|^p\,d\mathbb{P}=0,$$

so we can inductively find an $M_1 > M_0$ such that

$$\int_{\{|X_n|\geqslant M_1\}}|X_n|^p\,d\mathbb{P}<\varepsilon \quad \text{for } n=1,2,\ldots,N_0.$$

In conclusion,

$$\sup_{n\in\mathbb{N}} \int_{\{X_n|\geqslant M_1\}} |X_n|^p \, d\mathbb{P} \leqslant \varepsilon.$$

That is,

$$\lim_{M\to\infty} \left(\sup_{n\in\mathbb{N}} \int_{\{|X_n|\geqslant M\}} |X_n|^p \, d\mathbb{P} \right) = 0.$$

(ii) $\Rightarrow$(i): Given $\varepsilon > 0$, we can find $M > 0$ such that

$$\sup_{n\in\mathbb{N}} \int_{\{|X_n|>M\}} |X_n|^p \, d\mathbb{P} < \varepsilon.$$

Let $g := M + 2|X|$. Using a similar argument, we obtain

$$\int_{\{|X_n|\leqslant g\}} |X_n|^p d\mathbb{P} \longrightarrow \int_\Omega |X|^p d\mathbb{P}.$$

Therefore,

$$\begin{aligned} \limsup_{n\to\infty} \left| \int_\Omega |X_n|^p d\mathbb{P} - \int_\Omega |X|^p d\mathbb{P} \right| &\leqslant \limsup_{n\to\infty} \int_{\{|X_n|>g\}} |X_n|^p d\mathbb{P} \\ &\leqslant \sup_{n\in\mathbb{N}} \int_{\{|X_n|>M\}} |X_n|^p d\mathbb{P}. \end{aligned}$$

As $\varepsilon \to 0$, we finally obtain (i). □

Corollary 4.3.4. *Let $X, X_1, X_2, \ldots$ be random variables, satisfying $X_n \xrightarrow[n\to\infty]{p} X$, and $(X_n)_{n\in\mathbb{N}}$ is uniformly p-integrable. Then $X \in \mathcal{L}^p$ and $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p} X$.*

Proof. We only need to prove that $X \in \mathcal{L}^p$. Since $X_n \xrightarrow[n\to\infty]{p} X$, there exists a subsequence $X_{n_k} \xrightarrow[k\to\infty]{a.s.} X$. By Fatou's lemma, we have

$$\mathbb{E}\left[|X|^p\right] = \mathbb{E}\left[\lim_{k\to\infty} |X_{n_k}|^p\right] \leqslant \liminf_{k\to\infty} \mathbb{E}\left[|X_{n_k}|^p\right] \leqslant \sup_{n\in\mathbb{N}} \mathbb{E}\left[|X_n|^p\right].$$

Since $(X_n)_{n\in\mathbb{N}}$ is uniformly p-integrable, there exists $M > 0$ such that

$$\sup_n \mathbb{E}\left[|X_n|^p ; |X_n| > M\right] \leqslant 1.$$

Thus,

$$\sup_n \mathbb{E}\left[|X_n|^p\right] \leqslant \sup_n \mathbb{E}\left[|X_n|^p ; |X_n| \leqslant M\right] + \sup_n \mathbb{E}\left[|X_n|^p ; |X_n| > M\right] \leqslant M^p + 1 < \infty.$$

Therefore,

$$\mathbb{E}\left[|X|^p\right] \leqslant \sup_n \mathbb{E}\left[|X_n|^p\right] < \infty.$$

□

Exercises

1. Prove: (i) Let $0 \leqslant X_n \leqslant X$ and $X \in \mathcal{L}^1$ If X_n converges to X in probability, then $X_n \to X$ converges in $\mathcal{L}^1$.

   (ii) Let $X_1, X_2, \cdots \in \mathcal{L}^1$, $\inf_n E(X_n) > -\infty$ and $X_n \downarrow X$ almost everywhere. Then, $X_n \to X$ converges in $\mathcal{L}^1$.

Chapter 5

Law of Large Numbers

5.1 Kolmogorov $0-1$ Law

Definition 5.1.1 (Tail σ-algebra)**.** Let $X_1, X_2, \ldots$ be random variables. Define

$$\mathcal{T}_n := \sigma\left(X_{n+1}, X_{n+2}, \ldots\right), \mathcal{T} := \bigcap_{n\in\mathbb{N}} \mathcal{T}_n.$$

The σ-algebra $\mathcal{T}$ is called the tail σ-algebra of the sequence $(X_n)_{n\in\mathbb{N}}$.

Example 5.1.2. The $\mathcal{T}$ contains many important events, such as:

$$F_1 := \left\{ \lim_{n\to\infty} X_n \text{ exists } \right\} := \left\{ \omega : \lim_{n\to\infty} X_n(\omega) \text{ exists } \right\}$$
$$F_2 := \left\{ \sum_{n=1}^{\infty} X_n \text{ converges } \right\}$$
$$F_3 := \left\{ \lim_{n\to\infty} \frac{X_1 + X_2 + \ldots + X_n}{n} \text{ exists } \right\}.$$

We will only prove that: $F_3 \in \mathcal{T}$. First, notice that

$$F_3 = F_3^{(n)} := \left\{ \omega : \lim_{k\to\infty} \frac{X_{n+1}(\omega) + \ldots + X_{n+k}(\omega)}{k} \text{ exists } \right\},$$

This is because

$$\begin{aligned}
\left\{ \lim_{k\to\infty} \frac{X_{n+1} + \ldots + X_{n+k}}{k} \text{ exists } \right\} &= \left\{ \lim_{k\to\infty} \frac{X_{n+1} + \ldots + X_{n+k}}{k} \cdot \frac{k}{k+n} \text{ exists } \right\} \\
&= \left\{ \lim_{k\to\infty} \frac{X_{n+1} + \ldots + X_{n+k}}{n+k} \text{ exists } \right\} \\
&= \left\{ \lim_{k\to\infty} \left(\frac{X_{n+1} + \ldots + X_{n+k}}{n+k} + \frac{X_1 + \cdots + X_n}{n+k} \right) \text{ exists } \right\}.
\end{aligned}$$

Let $Y_k^{(n)} = \frac{X_{n+1}+\cdots+X_{n+k}}{k}$, then $Y_k^{(n)}$ is $\mathcal{T}_n$-measurable, so

$$F_3^{(n)} = \left\{ \limsup_{k\to\infty} Y_k = \liminf_{k\to\infty} Y_k \right\} \cap \left\{ \liminf_{k\to\infty} Y_k \neq -\infty \right\} \cap \left\{ \limsup_{k\to\infty} Y_k \neq \infty \right\} \in \mathcal{T}_n.$$

Thus, for all $n \in \mathbb{N}$, $F_3 = F_3^{(n)} \in \mathcal{T}_n$, and therefore $F_3 \in \mathcal{T}$.

Theorem 5.1.3 (Kolmogorov $0-1$ Law)**.** *Let $X_1, X_2, \ldots$ be independent random variables, and let$\mathcal{T}$ be the tail σ-algebra of the sequence$(X_n)_{n\in\mathbb{N}}$. Then, $\mathcal{T}$ is $\mathbb{P}$-trivial, meaning that for any $A \in \mathcal{T}$, we have $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$.*

Proof. According to Exercise 2 in the chapter of Theorem 3.1, $\sigma(X_1, X_2, \ldots, X_n)$ is independent of $\mathcal{T}_n := \sigma(X_{n+1}, X_{n+2}, \ldots)$. If $A \in \mathcal{T}$, then $A \in \mathcal{T}_n$, and hence A is independent of $\sigma(X_1, \ldots, X_n)$ for all $n \in \mathbb{N}$. Let

$$\mathcal{C} := \bigcup_{n=1}^{\infty} \sigma(X_1, \ldots, X_n).$$

Then $\mathcal{C}$ is closed under intersections, and $\mathcal{C}$ is independent of $\{A\}$. By Property 3.1.6, we have that $\{\varnothing, A, A^c, \Omega\}$ is independent of $\sigma(\mathcal{C}) = \sigma(X_1, X_2, \ldots)$. Clearly, $A \in \sigma(X_1, X_2, \ldots)$ so A is independent of itself. Hence,

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2$$

Thus, $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$.

□

5.2 Law of Large Numbers

Theorem 5.2.1 (Weak Law of Large Numbers)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed random variables such that $\mathbb{E}\left[{X_i}^2\right] < \infty$. Then,*

$$\frac{1}{n}\sum_{j=1}^{n} X_j \xrightarrow[n\to\infty]{p} \mathbb{E}[X_1].$$

Proof. By the linearity of expectation,

$$\mathbb{E}\left[\frac{1}{n}\sum_{j=1}^{n} X_j\right] = \frac{1}{n} \cdot n \cdot \mathbb{E}[X_1] = \mathbb{E}[X_1].$$

By Chebyshev's inequality,

$$\begin{aligned}
\mathbb{P}\left(\left|\frac{1}{n}\sum_{j=1}^{n} X_j - \mathbb{E}[X_1]\right| > \varepsilon\right) &\leqslant 1/\varepsilon^2 \operatorname{Var}\left[\frac{1}{n}\sum_{j=1}^{n} X_j\right] \\
&= \frac{1}{\varepsilon^2 n^2} \operatorname{Var}\left[\sum_{j=1}^{n} X_j\right] \\
&= \frac{1}{\varepsilon^2 n^2} \cdot n \cdot \operatorname{Var}[X_1] \xrightarrow{n\to\infty} 0. \qquad (5.2.1)
\end{aligned}$$

□

Theorem 5.2.2 (Weierstrass Approximation Theorem)**.** *Let $f : [0,1] \longrightarrow \mathbb{R}$ be a continuous function. Then there exists a polynomial f_n of degree at most n such that*

$$\|f_n - f\|_\infty \xrightarrow{n\to\infty} 0,$$

where $\|f\|_\infty := \sup\{|f(x)| : x \in [0,1]\}$.

Proof. For $n \in \mathbb{N}$, define

$$f_n(x) := \sum_{k=0}^{n} f(\frac{k}{n})\binom{n}{k} x^k \cdot (1-x)^{n-k}, \quad x \in [0,1]$$

This polynomial is called the Bernstein polynomial of degree n.

Let any $\varepsilon > 0$ be given. Since f is continuous on $[0,1]$, it is uniformly continuous on $[0,1]$. Therefore, there exists $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon, \quad \forall x, y \in [0,1],\ |x-y| < \delta.$$

Now choose $p \in [0,1]$. Let $X_1, X_2, \ldots$ be independent random variables with $X_i \sim \mathrm{Ber}_p,\ i \in \mathbb{N}$. Then $S_n := X_1 + \ldots + X_n \sim B_{n,p}$ and

$$\mathbb{E}\left[f\left(\frac{S_n}{n}\right)\right] = \sum_{k=0}^{n} f(\frac{k}{n})\mathbb{P}\left(S_n = k\right) = f_n(p).$$

Thus, we obtain

$$\left|f\left(\frac{S_n}{n}\right) - f(p)\right| < \varepsilon + 2\|f\|_\infty \cdot \mathbb{1}_{\{|\frac{S_n}{n} - p| \geqslant \delta\}}$$

Note that $\mathrm{Var}\left(X_i\right) = p \cdot (1-p) \leqslant \frac{1}{4}$ and applying Theorem 5.2.1, for each $p \in [0,1]$, we have

$$\begin{aligned} |f_n(p) - f(p)| &\leqslant \mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right|\right] \\ &\leqslant \varepsilon + 2 \cdot \|f\|_\infty \cdot \mathbb{P}\left(\left|\frac{S_n}{n} - p\right| \geqslant \delta\right) \\ &\overset{(5.2.1)}{\leqslant} \varepsilon + \frac{\|f\|_\infty}{2\delta^2 n}. \end{aligned}$$

Thus, $\|f_n - f\|_\infty \xrightarrow{n\to\infty} 0$.

□

Theorem 5.2.3 (Strong Law of Large Numbers)**.** *Let $X_1, X_2 \ldots$ be independent and identically distributed, and suppose that $\mathbb{E}\left[|X_i|\right] < \infty$. Then,*

$$\frac{X_1 + X_2 + \ldots + X_n}{n} \xrightarrow[n\to\infty]{a.s.} \mathbb{E}\left[X_1\right].$$

Proof. We will later use the Martingale Convergence Theorem to prove this theorem.

□

Chapter 6

Central Limit Theorem

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space.

Objective: Central Limit Theorem (CLT): Let $(X_j)_{j\in\mathbb{N}} \subset L^2$ be a sequence of independent and identically distributed real random variables, satisfying $\mathbb{E}X_j = 0, \mathbb{V}X_j = 1$. Let $S_n := \sum_{i=1}^n X_i$. Then we have:

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} G \sim \nu_{0,1}(dx) = \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{x^2}{2}\right) dx$$

6.1 Characteristic Function

Definition 6.1.1 ($\mathbb{C}$-valued function integral)**.** (1) Let $f : \mathbb{R}^d \to \mathbb{C}$ be $\mathcal{B}(\mathbb{C})$-measurable, $f = u+iv$ and $u, v : \mathbb{R}^d \to \mathbb{R}, u = \operatorname{Re} f, v = \operatorname{Im} f$. We define

$$\int f d\mu := \int u d\mu + i \int v d\mu \quad \in \mathbb{C}$$

Note: f is $\mathcal{B}(\mathbb{C})$-measurable if and only if u, v are $\mathcal{B}(\mathbb{R})$-measurable. Therefore, (10.1) is well-defined.

(2) For all $p \geqslant 1$, we have:

$$\begin{aligned} L^p_{\mathbb{C}}(\mu) &= \left\{ f : \mathbb{R}^d \to \mathbb{C}; \int |f|^p d\mu < \infty \right\} \\ &= \left\{ f : \mathbb{R}^d \to \mathbb{C}; \operatorname{Re} f, \operatorname{Im} f \in L^p_{\mathbb{R}}(\mu) \right\} \\ &= L^p_{\mathbb{R}}(\mu) \oplus i L^p_{\mathbb{R}}(\mu) \end{aligned}$$

Remark 6.1.2. (1) Properties of the integral: Due to linearity, almost all properties are the same as those of real integrals. However, in general, the "order" statements are different.

(2) As an example, we prove the triangle inequality $\left|\int f d\mu\right| \leqslant \int |f| d\mu$: We have $\int f d\mu \in \mathbb{C}$, so there exists $\theta \in (0, 2\pi]$ such that $e^{i\theta} \cdot \int f d\mu = \left|\int f d\mu\right|$. Thus, we obtain:

$$\begin{aligned}\left|\int f d\mu\right| &= e^{i\theta} \cdot \int f d\mu = \int e^{i\theta} \cdot f d\mu \\ &= \operatorname{Re}\left(\int e^{\imath\vartheta} \cdot f d\mu\right) = \int \operatorname{Re}\left(e^{\imath\vartheta} \cdot f\right) d\mu \\ &\leqslant \int \left|e^{\imath\vartheta}\right| \cdot |f| d\mu = \int |f| d\mu\end{aligned}$$

(3) For all measurable functions $f, g : \mathbb{C} \to \mathbb{C}$ and independent random variables X, Y, we have: $\mathbb{E}(f(X) \cdot g(Y)) = \mathbb{E}(f(X)) \cdot \mathbb{E}(g(Y))$

Definition 6.1.3. Let $X : \Omega \to \mathbb{R}$ be a random variable, and denote its distribution by $\boldsymbol{P}_X$. We define

$$\varphi_X(t) = \mathbb{E}\left[e^{itX}\right] = \int_{\mathbb{R}} e^{ity} \boldsymbol{P}_X(dy), \quad t \in \mathbb{R}$$

as the characteristic function of X. Note that this function is always well-defined because

$$\left|e^{itx}\right| \leqslant 1$$

.

Theorem 6.1.4 (Properties of the Characteristic Function)**.** *Let $\varphi = \varphi_X$ be the characteristic function of a random variable X. Then:*

(i) $\varphi(0) = 1, |\varphi(t)| \leqslant 1, \varphi(-t) = \overline{\varphi(t)}$.

(ii) φ *is uniformly continuous on*$\mathbb{R}$.

(iii) $\varphi_{aX+b}(t) = e^{ibt}\, \varphi_x(at), \quad a, b \in \mathbb{R}$.

(iv) If X *and* Y *are independent, then* $\varphi_{X+Y}(t) = \varphi_X(t) \cdot \varphi_Y(t)$.

Proof. (i) Trivial.

(ii) We have

$$\begin{aligned}|\varphi(t+h) - \varphi(t)| &\leqslant \left|\mathbb{E}e^{i(t+h)X} - \mathbb{E}e^{itX}\right| \leqslant \mathbb{E}\left|e^{itX}\left(e^{ihX} - 1\right)\right| \\ &\leqslant \mathbb{E}\left|e^{ihx} = 1\right|.\end{aligned}$$

As $h \to 0$, then$\left|e^{ihx} - 1\right| \to 0$. By the dominated convergence theorem,

$$\mathbb{E}|e^{ih} - 1| \to 0, \text{ as } h \to 0.$$

For any given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\mathbb{E}|e^{ihx} - 1| < \varepsilon, \quad |h| < \delta.$$

Therefore, $|\varphi(t+h) - \varphi(t)| < \varepsilon$, $|h| < \delta$.

(iii) $\mathbb{E}\left[e^{it(aX+bY)}\right] = e^{itb}\mathbb{E}\left[e^{iatx}\right] = e^{itb}\varphi_X(at)$

(v) $\mathbb{E}\left[e^{it(X+Y)}\right] = \mathbb{E}\left[e^{itx} \cdot e^{itY}\right] = \mathbb{E}\left[e^{itx}\right] \cdot E\left[e^{itY}\right]$. □

Theorem 6.1.5 (Characteristic Function and Moments)**.** *Let X be a random variable taking values in $\mathbb{R}$.*

(1). If $X \in \mathcal{L}^p$, then φ_X is p times continuously differentiable, and for $k = 0, \ldots, p$, we have $\varphi_X^{(k)}(t) = \mathbf{E}\left[(iX)^k e^{itX}\right]$. In particular, $\varphi_X^{(k)}(0) = i^k \mathbf{E}\left[X^k\right]$.

(2). Specifically, if $X \in \mathcal{L}^2$, then $\varphi_X(t) = 1 + it\mathbf{E}[X] - \frac{t^2}{2}\mathbf{E}\left[X^2\right] + \varepsilon(t)t^2$ where $\varepsilon(t) \xrightarrow{t\to 0} 0$.

Proof. 1. Since $|X|^p$ is integrable, $|X|^p \vee 1$ is also integrable. Thus, all $|X|^k$ have an integrable upper bound, and the expression on the right-hand side exists. Since the statement is obviously true when $k = 0$, assume it holds for $k < n$. Then, $\left|\frac{d^{k+1}}{dt^{k+1}} e^{itX}\right| = \lim_{h\to 0}\left|\frac{(iX)^k e^{i(t+h)X} - (iX)^k e^{itX}}{h}\right| \leqslant \left|X^k \frac{e^{ihX}-1}{h}\right| \leqslant \left|X^{k+1}\right|$. By dominated convergence, we can interchange the derivative and the integral, yielding $\varphi_X^{(k+1)}(t) = \mathbf{E}\left[\frac{d}{dt}(iX)^k e^{itX}\right] = \mathbf{E}\left[(iX)^{k+1} e^{itX}\right]$.

The continuity of derivatives also follows from dominated convergence.

2. For the estimation, we need the Taylor expansion of φ_X and the remainder term. We have $e^{itX} = 1 + itX - \frac{t^2X^2}{2}\left(\cos(\theta_1 tX) + i\sin(\theta_2 tX)\right)$ where θ_1, θ_2 are random variables and $|\theta_1|, |\theta_2| \leqslant 1$. Therefore, we obtain $\varphi_X(t) = 1 + it\mathbf{E}[X] - \frac{t^2}{2}\mathbf{E}\left[X^2\right] + \varepsilon(t)t^2$ where $2\varepsilon(t) = \mathbf{E}\left[X^2\left(1 - \cos(\theta_1 tX) + i\sin(\theta_2 tX)\right)\right] \xrightarrow{t\to 0} 0$, which follows from dominated convergence. □

Theorem 6.1.6 (Lévy, Inversion Formula)**.** *Let φ be the characteristic function of a random variable X, and μ be the distribution of X. Then for all $a < b$, $a, b \in \mathbb{R}$, we have: For any $a < b$, $a, b \in \mathbb{R}$:*

$$\lim_{T\to\infty} \frac{1}{2\pi} \cdot \int_{-T}^{T} \frac{e^{-i\theta a} - e^{-i\theta b}}{i\theta} \varphi(\theta) d\theta = \frac{1}{2}\mu\{a\} + \mu(a,\ b) + \frac{1}{2}\mu\{b\}$$

Proof. See the appendix. □

Theorem 6.1.7. *Let X, Y be random variables. If $\varphi_X = \varphi_Y$, then $\mathbb{P}_X = \mathbb{P}_Y$.*

Proof. Let $\mu_1 = \mathbb{P}_X, \mu_2 = \mathbb{P}_Y$. By the inversion formula, for $a < b$ we have

$$\frac{1}{2}\mu_1(\{a\}) + \mu_1(a,b) + \frac{1}{2}\mu_1(\{b\}) = \frac{1}{2}\mu_2(\{a\}) + \mu_2(a,b) + \frac{1}{2}\mu_2(\{b\})$$

Since $\frac{1}{2}\mu_1(\{a\}) + \mu_1(a,b) \leqslant \mu_1([a,b)) \downarrow 0$, when $a \uparrow b$.

In the above equation, let $a \uparrow b$, we get

$$\mu_1(\{b\}) = \mu_2(\{b\})$$

Therefore, $\mu_1([a,b]) = \mu_2([a,b]), \quad a < b$. Since $\mathcal{E} := \{(a,b] : a < b, a, b \in \mathbb{R}\}$ is closed under intersection and $\sigma(\mathcal{E}) = \mathcal{B}(\mathbb{R})$, by the uniqueness of measures, we conclude that $\mu_1 = \mu_2$. □

6.2 Weak Convergence of Probability Measures

Definition 6.2.1 (Weak Convergence and Convergence in Distribution)**.** (i) We denote by $\mathcal{P}(\mathbb{R})$ the set of probability measures on $\mathbb{R}$. Moreover, $\mathcal{Z}_b(\mathbb{R})$ is the set of bounded continuous functions on $\mathbb{R}$.

(ii) If a sequence $\mu_1, \mu_2, \ldots \in \mathcal{P}(\mathbb{R})$ satisfies

$$\int_{\mathbb{R}} g d\mu_n \longrightarrow \int_{\mathbb{R}} g d\mu \quad \text{for all} \quad g \in \mathcal{Z}_b(\mathbb{R})$$

then we say that the sequence weakly converges to $\mu \in \mathcal{P}(\mathbb{R})$, and write $\mu_n \xrightarrow{w} \mu$.

(iii) Let $X, X_1, X_2, \ldots$ be random variables defined on probability spaces $(\Omega, \mathcal{F}, \mathbb{P}), (\Omega_1, \mathcal{F}_1, P_1), (\Omega_2, \mathcal{F}_2, P_2), \ldots$, and let $\mu, \mu_1, \mu_2, \ldots$ be the distributions of $X, X_1, X_2, \ldots$. If $\mu_n \xrightarrow{w} \mu$, then we say that X_n converges in distribution to X, and write $X_n \xrightarrow{d} X$.

Remark 6.2.2. Convergence in distribution is not a property of the random variables themselves, but rather a property of their distributions.

Theorem 6.2.3. *Let $X, X_1, X_2, \ldots$ be random variables defined on probability spaces $(\Omega, \mathcal{A}, P), (\Omega_1, \mathcal{A}_1, P_1), \ldots$, and let $F, F_1, F_2, \ldots$ be the distribution functions of $X, X_1, X_2, \ldots$. The following two statements are equivalent:*

(1) X_n converges in distribution to X (denoted $X_n \xrightarrow{d} X$).

(2) For every continuous point $y \in \mathbb{R}$ of F, we have $F_n(y) \to F(y)$.

Proof. "$(1) \Rightarrow (2)$": Let F be continuous at point x. For any $\varepsilon > 0$, choose $\delta > 0$ such that for all y satisfying $|y - x| < \delta$, we have $|F(y) - F(x)| < \varepsilon$.

Let g be a continuous function such that $0 \leqslant g \leqslant 1$, and $g(z) = 1$ for $z \in (-\infty, x]$ and $g(z) = 0$ for $z \in [x + \delta, \infty)$. Then

$$F_n(x) \leqslant \mathbb{E}[g(X_n)] \to \mathbb{E}[g(X)] \leqslant F(x + \delta) \leqslant F(x) + \varepsilon.$$

Similarly, let h be a continuous function such that $0 \leqslant h \leqslant 1$, and $h(z) = 1$ for $z \in (-\infty, x - \delta]$ and $h(z) = 0$ for $z \in [x, \infty)$. Then

$$F_n(x) \geqslant \mathbb{E}[h(X_n)] \to \mathbb{E}[h(X)] \geqslant F(x - \delta) \geqslant F(x) - \varepsilon.$$

Since ε is arbitrary, we have $F_n(x) \to F(x)$.

"$(2) \Rightarrow (1)$": Let $C := \{x \in \mathbb{R} : F \text{ is continuous at } x\}$, and$D := \mathbb{R} \backslash C$. Then D is countable. From the definition, for $a, b \in C$ with $a < b$, we have

$$\mathbb{E}[\mathbb{1}_{(a,b]}(X_n)] = F_n(b) - F_n(a) \to F(b) - F(a) = \mathbb{E}[\mathbb{1}_{(a,b]}(X)].$$

Now, let g be a continuous function such that $|g| \leqslant M$ for some $M > 0$. For any $\varepsilon > 0$, choose $a_0, b_0 \in C$ such that $F(a_0) < \varepsilon$ and $F(b_0) > 1 - \varepsilon$. Since g is uniformly continuous on $[a_0, b_0]$, we

can find a sequence of continuous points $a_0 = c_1 < \ldots < c_k = b_0$ such that for all x satisfying $c_{i-1} \leqslant x \leqslant c_i$, we have $|g(x) - g(c_i)| < \varepsilon$.

Then, we have

$$\begin{aligned}\mathbb{E}[g(X_n)] &= \sum_{i=2}^{k} \mathbb{E}[g(X_n); c_{i-1} < X_n \leqslant c_i] + \mathbb{E}[g(X_n); X_n \leqslant c_1] + \mathbb{E}[g(X_n); X_n > c_k] \\ &\leqslant 2\varepsilon M + \sum_{i=2}^{k} \mathbb{E}[(g(c_i) + \varepsilon) 1_{(c_{i-1}, c_i]}(X_n)].\end{aligned}$$

$$\begin{aligned}\limsup_n \mathbb{E}\left[g\left(X_n\right)\right] &\leqslant \sum_{i=2}^{k} \mathbb{E}\left[\left(g\left(c_i\right) + \varepsilon\right) \cdot \mathbb{1}_{(c_{i-1}, c_i]}(X)\right] + 2\varepsilon \cdot M \\ &\leqslant \sum_{i=2}^{k} \mathbb{E}\left[(g(X) + 2\varepsilon) \cdot \mathbb{1}_{(c_{i-1}, c_i]}(X)\right] + 2\varepsilon \cdot M \\ &\leqslant \sum_{i=2}^{k} \mathbb{E}\left[g(X) \cdot \mathbb{1}_{(c_{i-1}, c_i]}(X)\right] + 2\varepsilon \cdot (M+1) \\ &\leqslant \mathbb{E}[g(X)] - \mathbb{E}\left[g(X); X \leqslant c_1\right] - \mathbb{E}\left[g(X); X > c_k\right] \\ &\quad + 2\varepsilon \cdot (M+1) \\ &\leqslant \mathbb{E}[g(X)] + 2\varepsilon \cdot (M+1).\end{aligned}$$

Let $\varepsilon \to 0$, $\quad \limsup_n \mathbb{E}\left[g\left(X_n\right)\right] \leqslant \mathbb{E}[g(X)]$. Similarly, for the function $-g$, we can prove $\liminf \mathbb{E}[g(X_n)] \geqslant \mathbb{E}[g(X)]$. □

Definition 6.2.4 (Tightness)**.** (1) A family of probability measures $(\mu_i)_{i \in I}$ in $\mathcal{P}(\mathbb{R})$ is said to be tight if for any $\varepsilon > 0$, there exists $K > 0$ such that

$$\mu_i([-K, K]) > 1 - \varepsilon, \quad \forall i \in I.$$

(2) A family of random variable $(X_i)_{i \in I}$ is said to be tight if their distributions $(\mathbb{P}_{X_n})_{i \in I}$ are tight.

Theorem 6.2.5 (Helly)**.** *Let $\mu_1, \mu_2, \ldots \in \mathcal{P}(\mathbb{R})$, and the sequence $(\mu_n)_{n \in \mathbb{N}}$ be tight. Then there exists a subsequence $(\mu_{n_k})_{k \in \mathbb{N}}$ that weakly converges to some $\mu \in \mathcal{P}(\mathbb{R})$, i.e.,*

$$\mu_{n_k} \xrightarrow{w} \mu \quad \textit{when} \quad k \to \infty.$$

Proof. The proof is in three steps: (1) Let $F_1, F_2, \ldots$ be the distribution functions of $\mu_1, \mu_2, \ldots$, i.e.,

$$F_n(x) := \mu_n((-\infty, x)); x \in \mathbb{R}.$$

Let q_k be an enumeration of the rational numbers. Since $F_n\left(q_k\right) \in [0, 1]$, any subsequence has a further subsequence that converges. Using the diagonalization argument, we make $F_{n_j}\left(q_k\right)$ converge for each q_k and define the limit as $F\left(q_k\right)$. The function F is non-decreasing, and define $F(x) = \inf_{q_k \geqslant x} F\left(q_k\right)$. Therefore, F is right-continuous and non-decreasing.

(2) If x is a continuous point of F and $\varepsilon > 0$, then there exist rationals r and s such that $r < x < s$ and $F(s) - \varepsilon < F(x) < F(r) + \varepsilon$. Then, we have

$$F_{n_j}(x) \geqslant F_{n_j}(r) \to F(r) > F(x) - \varepsilon$$

and

$$F_{n_j}(x) \leqslant F_{n_j}(s) \to F(s) < F(x) + \varepsilon.$$

Since ε is arbitrary, $F_{n_j}(x) \to F(x)$.

(3) Since F_n is tight, there exists M such that $F_n(-M) < \varepsilon$. Then, $F(-M) \leqslant \varepsilon$, which implies $\lim_{x\to-\infty} F(x) = 0$. Similarly, it can be shown that $\lim_{x\to\infty} F(x) = 1$. Thus, F is indeed a distribution function.

From (1), (3), and Proposition 2.0.4, we conclude that there exists a $\mu \in \mathcal{P}(\mathbb{R})$ such that $\mu((-\infty, x]) = F(x)$ for all $x \in \mathbb{R}$. From (2) and Theorem 6.2.3, we conclude that $\mu_n \xrightarrow[k\to\infty]{w} \mu$. □

Proposition 6.2.6. *Suppose there exists a function $\varphi : [0, \infty) \to [0, \infty)$, which is increasing, and $x \to \infty$ as $\varphi(x) \to \infty$. If $c = \sup_n \mathbb{E}\varphi(|X_n|) < \infty$, then X_n is tight.*

Proof. Let $\varepsilon > 0$. Choose M such that for $x > M$, $\varphi(x) \geqslant c/\varepsilon$. Then

$$\mathbb{P}(|X_n| > M) \leqslant \int \frac{\varphi(|X_n|)}{c/\varepsilon} \mathbb{1}_{(|X_n|>M)} d\mathbb{P} \leqslant \frac{\varepsilon}{c}\mathbb{E}\varphi(|X_n|) \leqslant \varepsilon$$

□

Notation: For $\mu \in \mathcal{P}(\mathbb{R})$, define its Fourier transform $\hat{\mu}(t) := \int_{\mathbb{R}} e^{itx}\mu(dx)$.

Theorem 6.2.7 (Lévy Continuity Theorem)**.** *Let $(\mu_n)_{n\in\mathbb{N}} \in \mathcal{P}(\mathbb{R})$.*

(i) $\mu_n \xrightarrow{w} \mu, \mu \in \mathcal{P}(\mathbb{R}) \Rightarrow \hat{\mu}_n(t) \to \hat{\mu}(t)$, and the convergence is uniform on any finite interval.

(ii) If $\hat{\mu}_n(t) \to \varphi(t)$, $\forall t \in \mathbb{R}$ and φ is continuous at 0, then: (1) $\exists \mu \in \mathcal{P}(\mathbb{R})$ such that$\varphi = \hat{\mu}$; (2) $\mu_n \xrightarrow[n\to\infty]{W} \mu$.

Proof. See appendix. □

6.3 Central Limit Theorem

Lemma 6.3.1. *Let $(c_n)_{n\in\mathbb{N}} \subset \mathbb{C}$ and $c_n \to c \in \mathbb{C}$ as $n \to \infty$. Then we have:*

$$\lim_{n\to\infty} \left(1 + \frac{c_n}{n}\right)^n = e^c.$$

Proof. See appendix. □

Theorem 6.3.2 (Central Limit Theorem)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed random variables, where $\mathbb{E}(X_1) = 0, \mathrm{Var}(X_1) = 1$. Then we have:*

$$G_n := \frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}} \xrightarrow[n\to\infty]{w} G \sim N(0,1)$$

where $N(0,1)$ denotes the normal distribution with mean 0 and variance 1.

Proof. Since $\mathrm{Var}(X_1) = 1$, we have $\mathbb{E}[X_1^2] = 1 < \infty$. By Theorem 6.1.5, we know that the characteristic function of X_1 is $\varphi_{X_1}(t) = 1 - \frac{t^2}{2} + \epsilon(t)\cdot t^2$, where $\epsilon(t) \xrightarrow{t\to 0} 0$.

Therefore, the characteristic function ofG_n is:

$$\varphi_{G_n}(t) = \left(\varphi_{X_1}\left(\frac{t}{\sqrt{n}}\right)\right)^n = \left[1 - \frac{t^2}{2n} + \epsilon\left(\frac{t}{\sqrt{n}}\right)\cdot\frac{t^2}{n}\right]^n$$

$$= \left(1 - \frac{t^2}{2n}\left(1 - 2\epsilon\left(\frac{t}{\sqrt{n}}\right)\right)\right)^n.$$

As $n \to \infty$, $-\frac{t^2}{2}\left(1 - 2\epsilon\left(\frac{t}{\sqrt{n}}\right)\right) \to -\frac{t^2}{2}$.

By Lemma 6.3.1, we have $\varphi_{G_n}(t) \xrightarrow{n\to\infty} e^{-\frac{t^2}{2}}$.

By Lévy's continuity theorem, we have $G_n \xrightarrow[n\to\infty]{w} G \sim N(0,1)$, where the probability density function of $N(0,1)$ is:

$$\frac{1}{\sqrt{2\pi}}\cdot\exp\left(-\frac{x^2}{2}\right)dx.$$

Since $\int_{\mathbb{R}} e^{itx}\cdot\frac{1}{\sqrt{2\pi}}\cdot e^{-\frac{x^2}{2}}dx = e^{-\frac{t^2}{2}}$. □